

Started on	Monday, 29 November 2021, 1:28 PM
State	Finished
Completed on	Monday, 29 November 2021, 1:28 PM
Time taken	11 secs
Marks	0/30
Grade	0 out of 100



Question 1

Not answered

Marked out of 1

In the below graph, the median survival time is given by

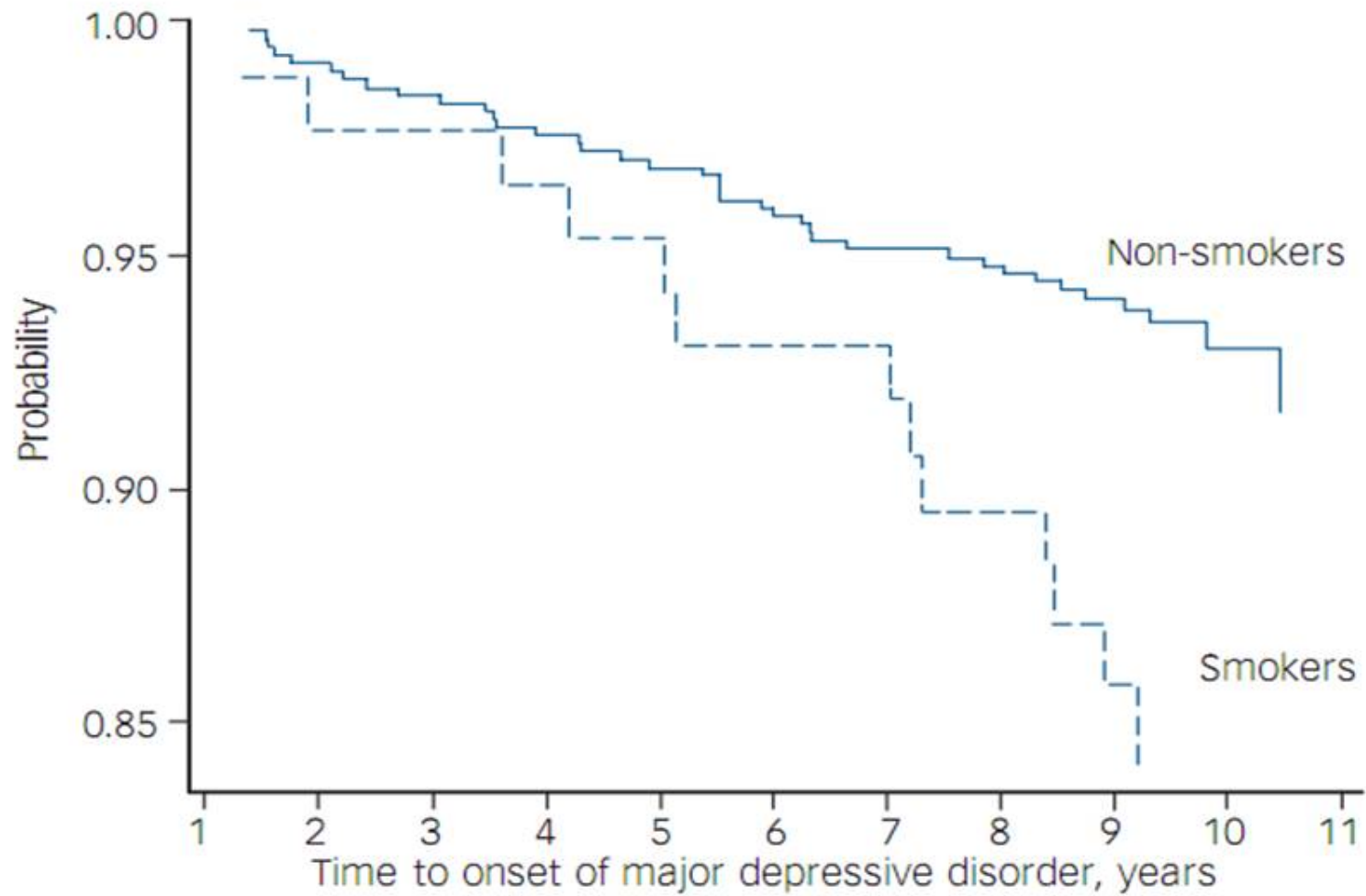


Fig. 2 Survival plot (Kaplan–Meier) showing the probability of remaining free of *de novo* major depressive disorder over a 10-year period for smokers and non-smokers at baseline.

Select one:

- ☐ 5 years
- ☐ 9 years
- ☐ 7 years
- ☐ 10 years
- ☐ Cannot be determined

Your answer is incorrect.

This survival curve is right censored to an extent that we have not been shown the time taken to be reduced to 50% surviving proportion. In this case, we cannot determine median survival time directly. British Journal of Psychiatry (2008) 193, 322-326

The correct answer is: Cannot be determined

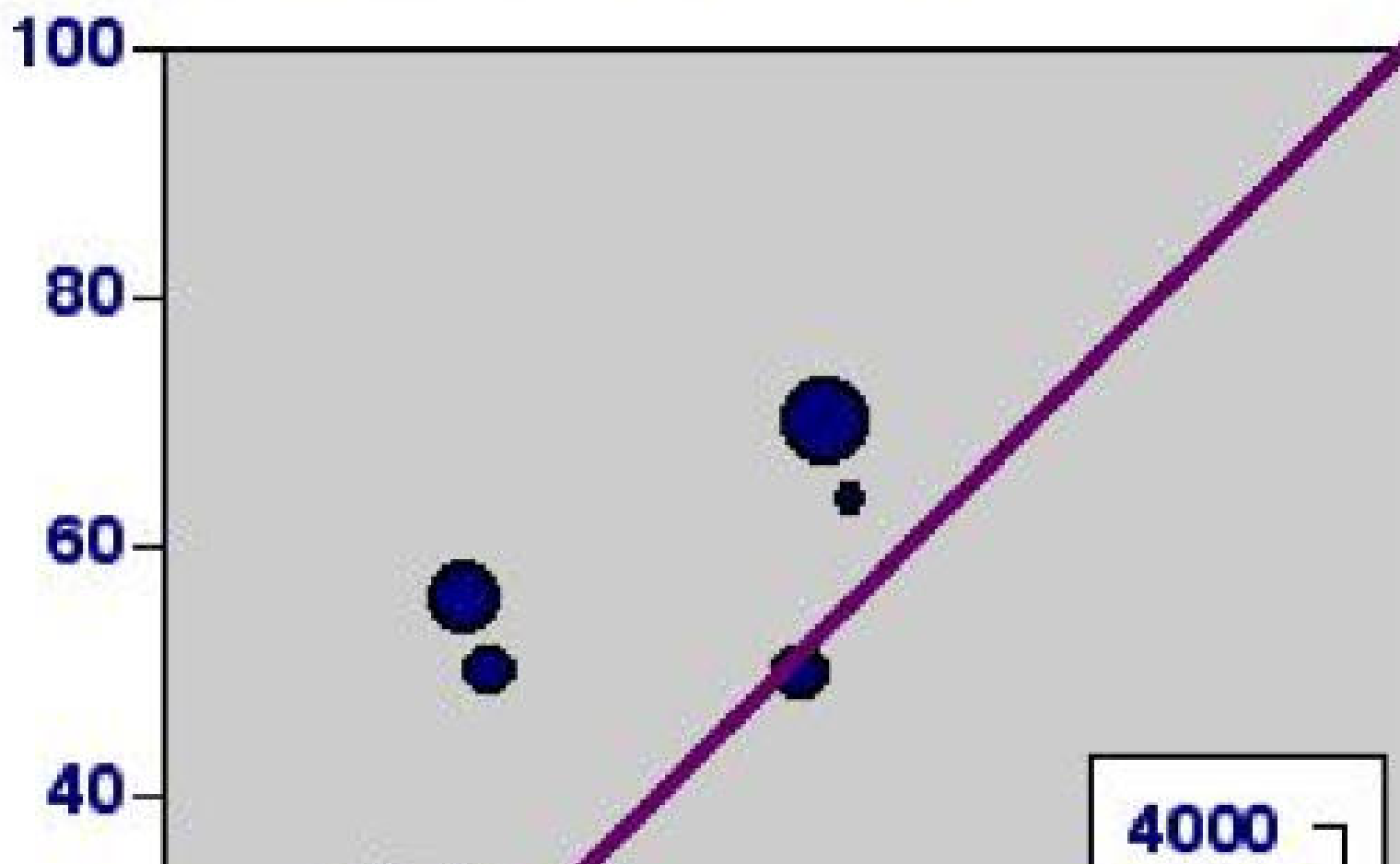
Question 2

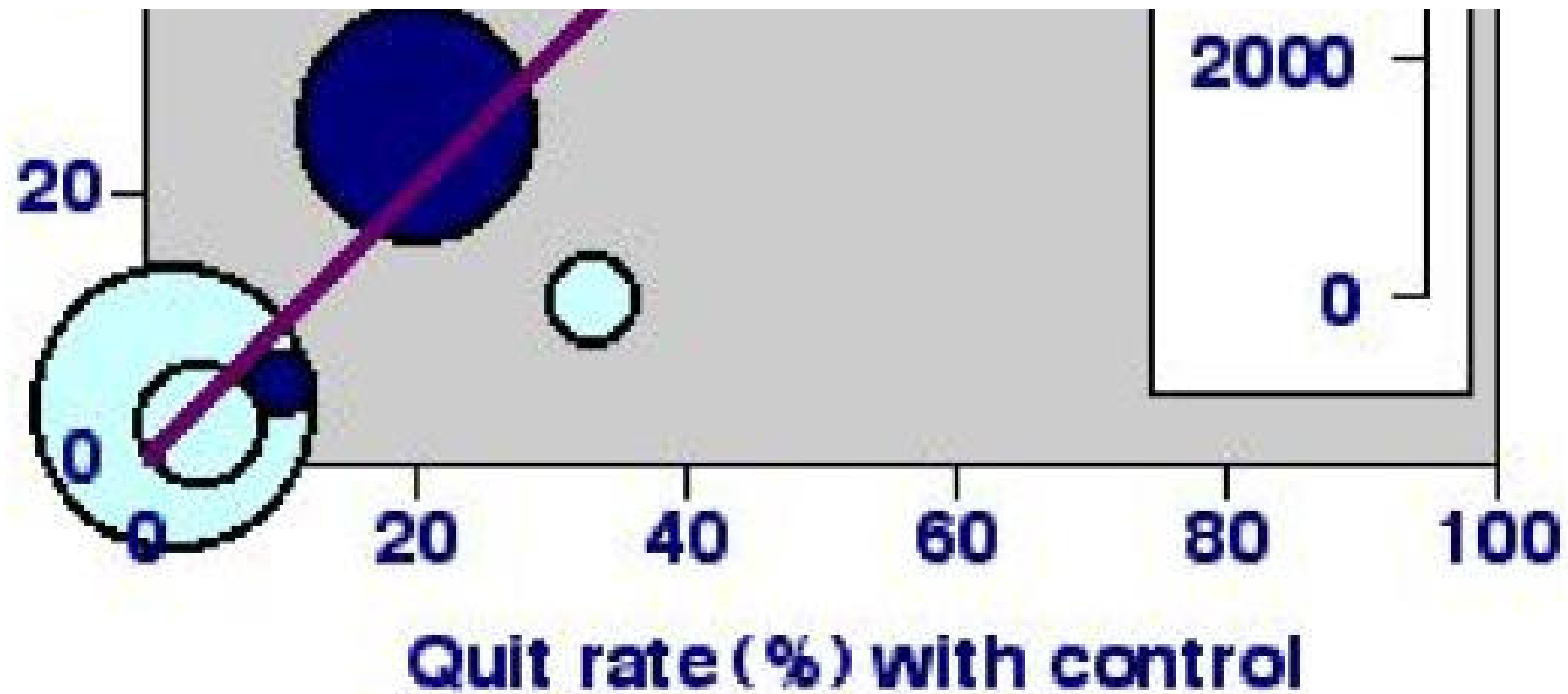
Not answered

Marked out of 1

In the below graph, the number of studies that demonstrate clear benefit for intervention compared to control irrespective of sample size is

Quit rate (%) with intervention





Select one:

- ☐ 4
- ☐ 3
- ☐ 2
- ☐ 8
- ☐ 7

Your answer is incorrect.

As there are four circles clearly above the line of equality, the answer is 4.

The correct answer is: 4

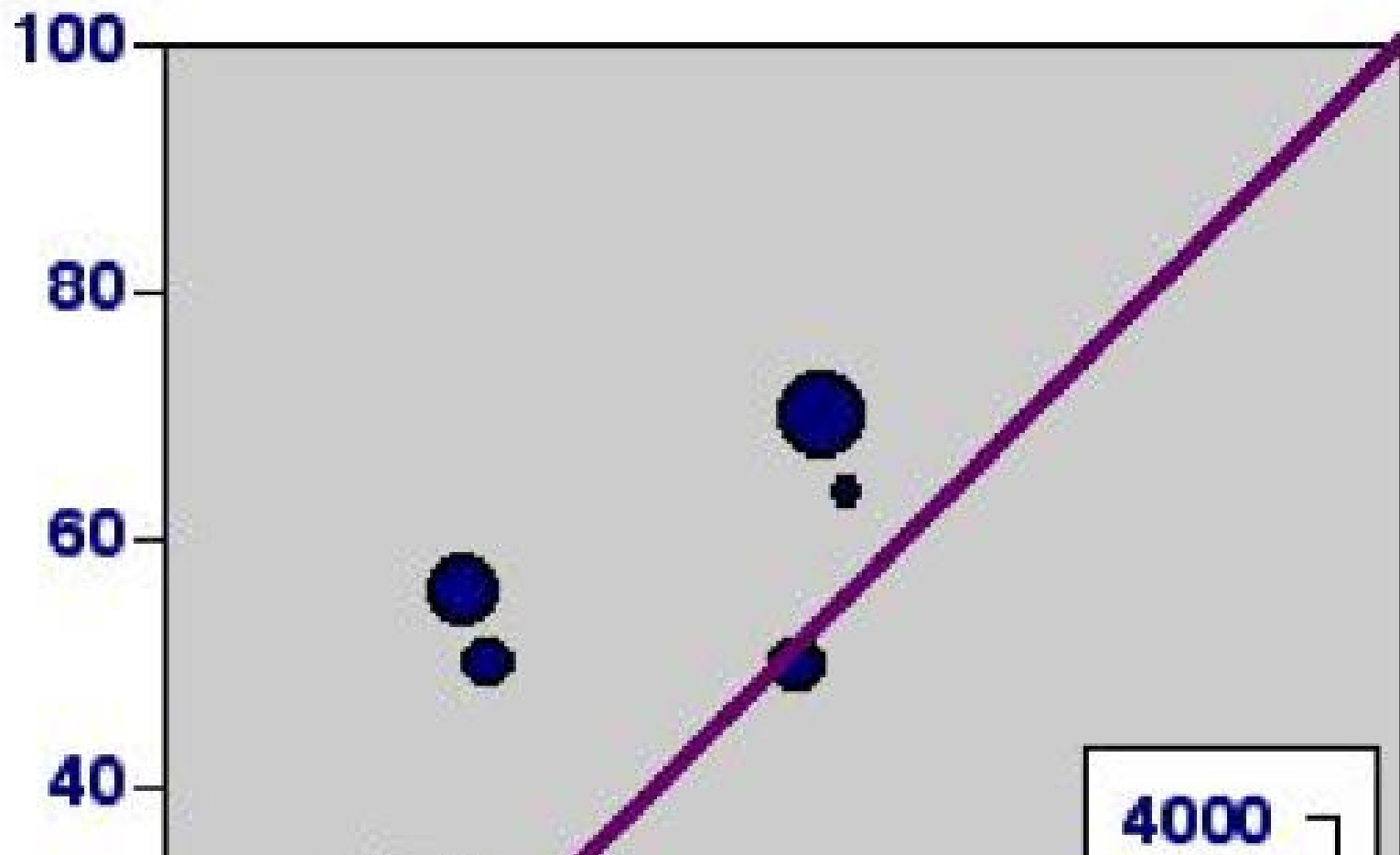
Question 3

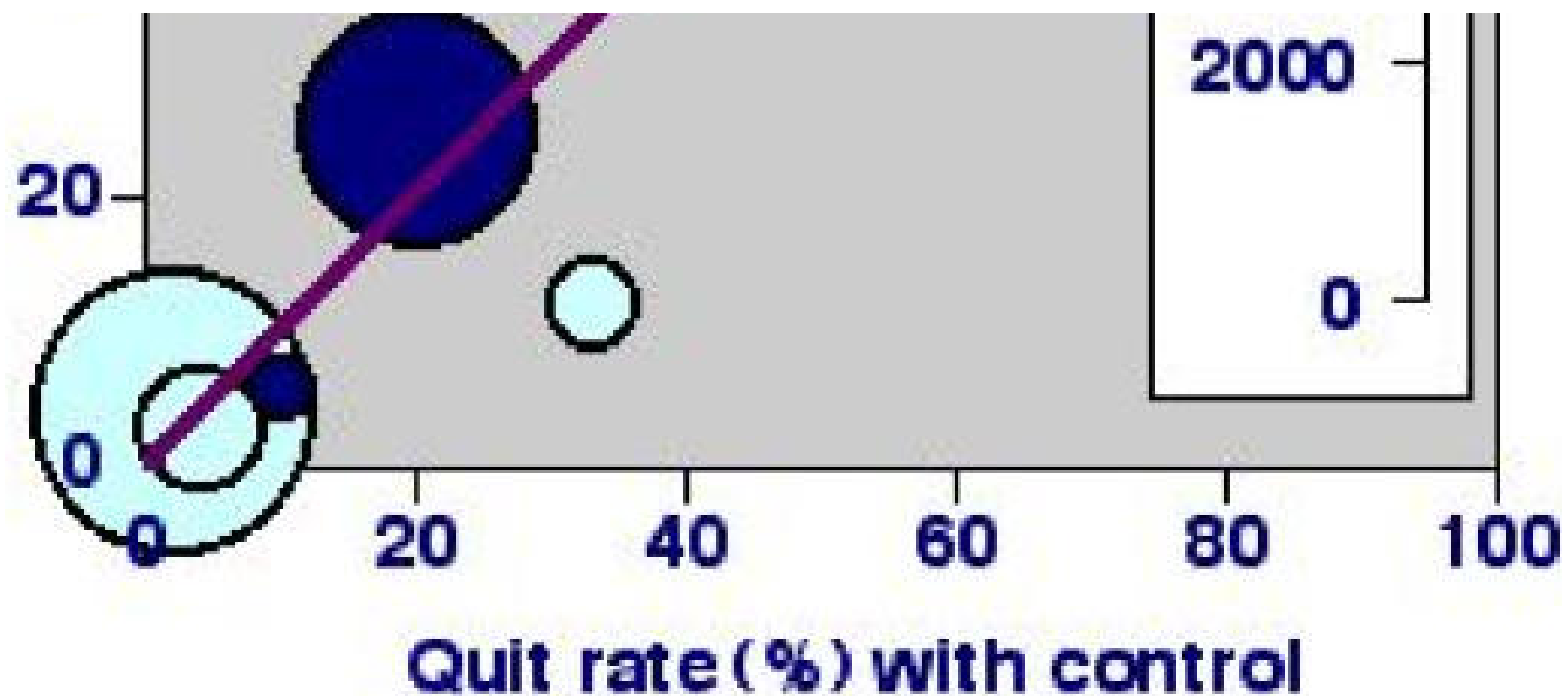
Not answered

Marked out of 1

In the below graph, the size of the circles shown corresponds to

Quit rate (%) with intervention





Select one:

- ☐ Reliability of a study result
- ☐ sample size in individual studies
- ☐ magnitude of treatment effect
- ☐ variance observed within a study
- ☐ confidence interval width

Your answer is incorrect.

Generally L'Abbe plots have the symbols appropriate to the size of the trials. We can also use different colour or shades to show clinical heterogeneity. Here Dark blue symbols indicate studies in a hospital setting and light blue symbols those in a primary care setting. Moore et al. BMC Medical Research Methodology 2002 2:2

The correct answer is: sample size in individual studies



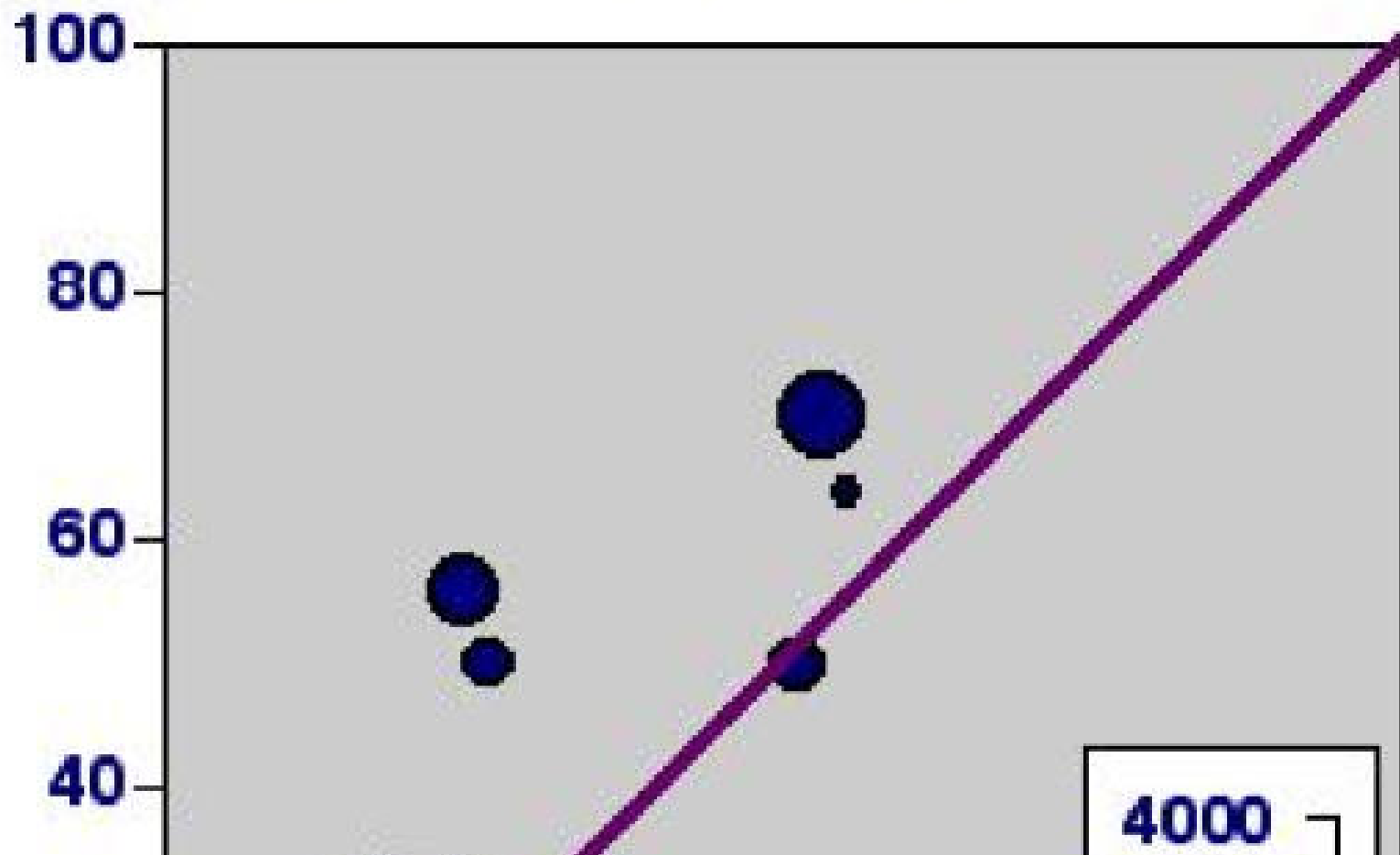
Question 4

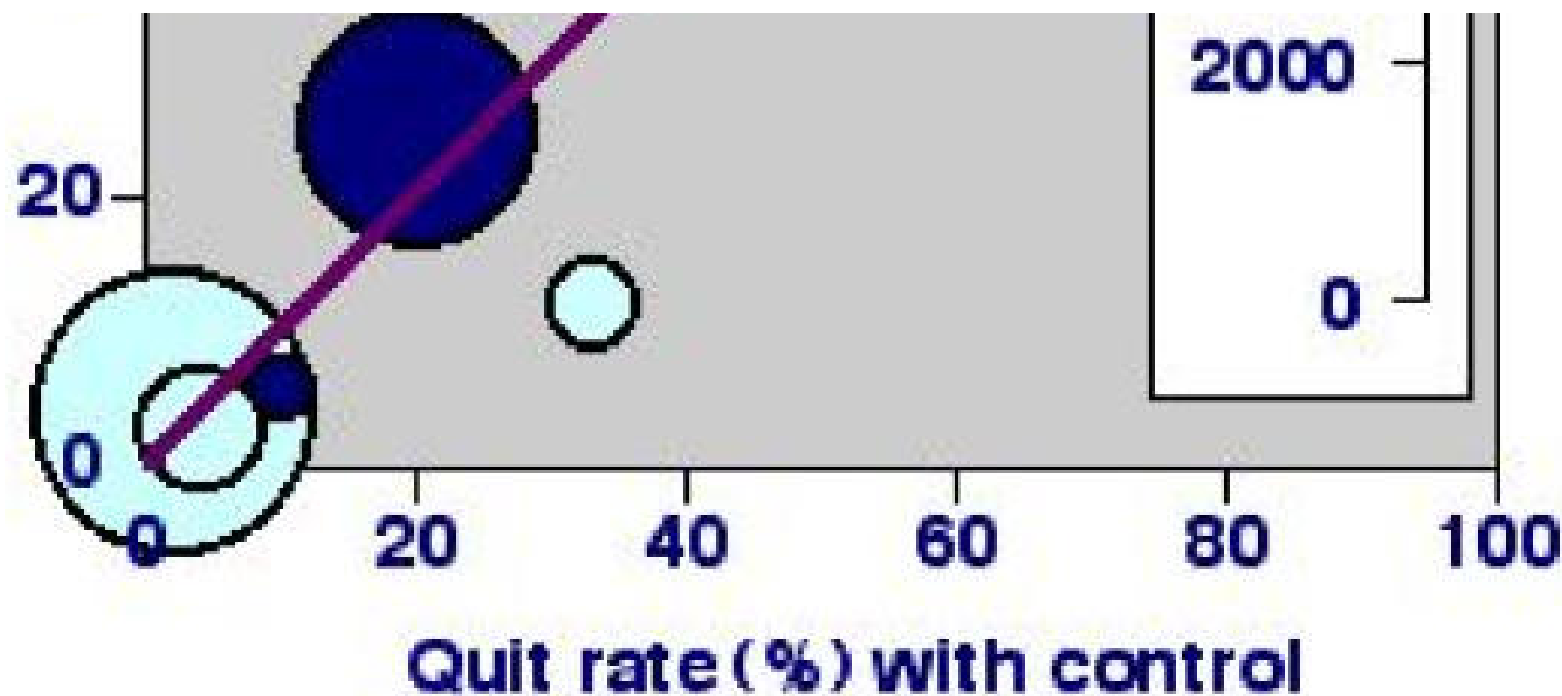
Not answered

Marked out of 1

In the below graph, each circle shown represents

Quit rate (%) with intervention





Select one:

- ☐ individual studies in a meta-analysis
- ☐ individual studies with overall favourable outcome
- ☐ studies influenced by publication bias
- ☐ different centres in a multicentre mega RCT
- ☐ each subject in a trial

Your answer is incorrect.

L'Abbe plot is a simple plot, but it can yield a comprehensive qualitative view of the data. Trials in which the experimental treatment proves better than the control ($EER > CER$) will be in the upper left of the plot, between the y axis and the line of equality. If experimental is no better than control then the point will fall on the line of equality ($EER = CER$), and if control is better than experimental then the point will be

in the lower right of the plot, between the x-axis and the line of equality ($EER < CER$) [Excerpt retrieved from <http://www.medicine.ox.ac.uk/bandolier/booth/glossary/labbe.html>]. Here each circle represents an individual study.

The correct answer is: individual studies in a meta-analysis



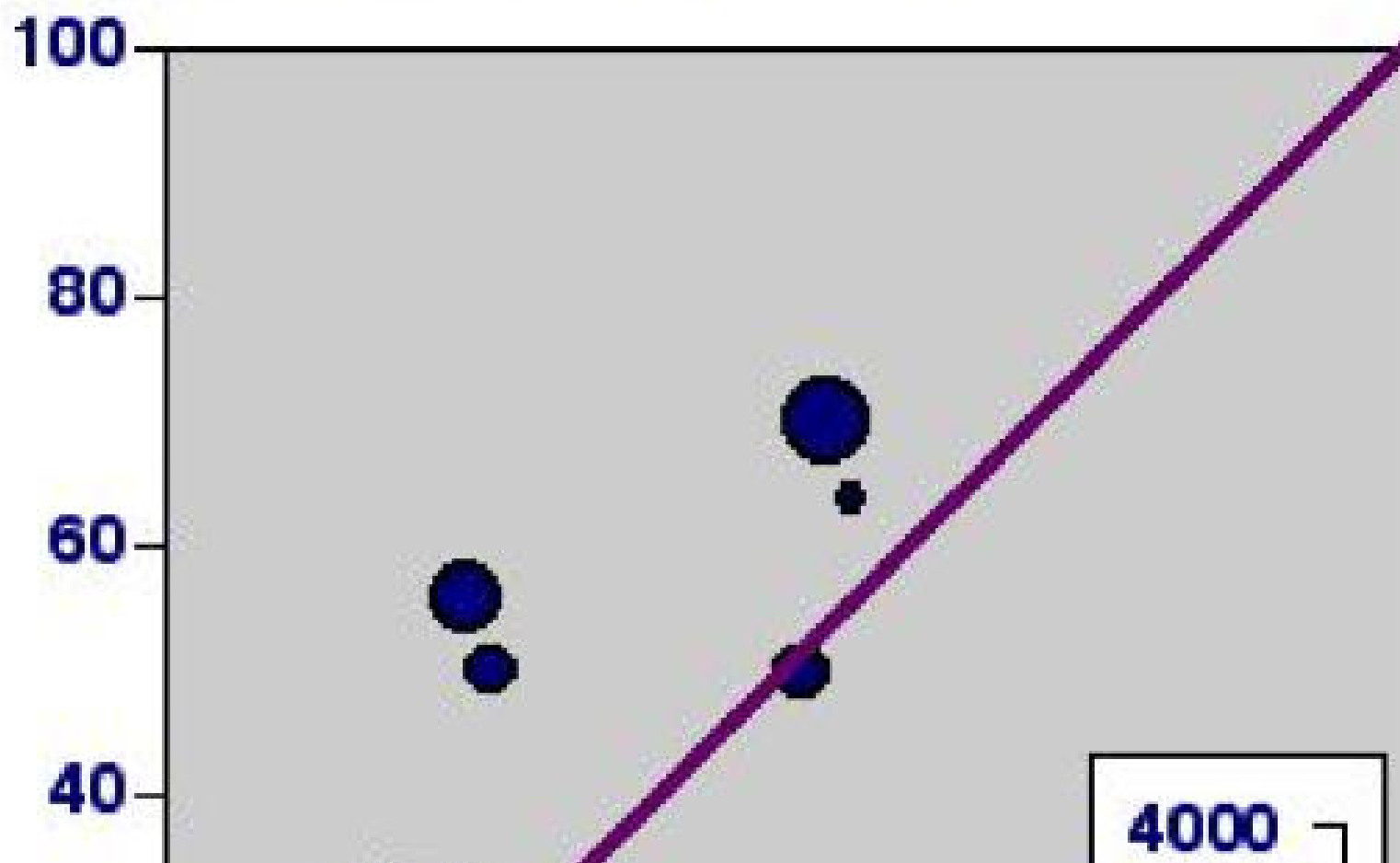
Question 5

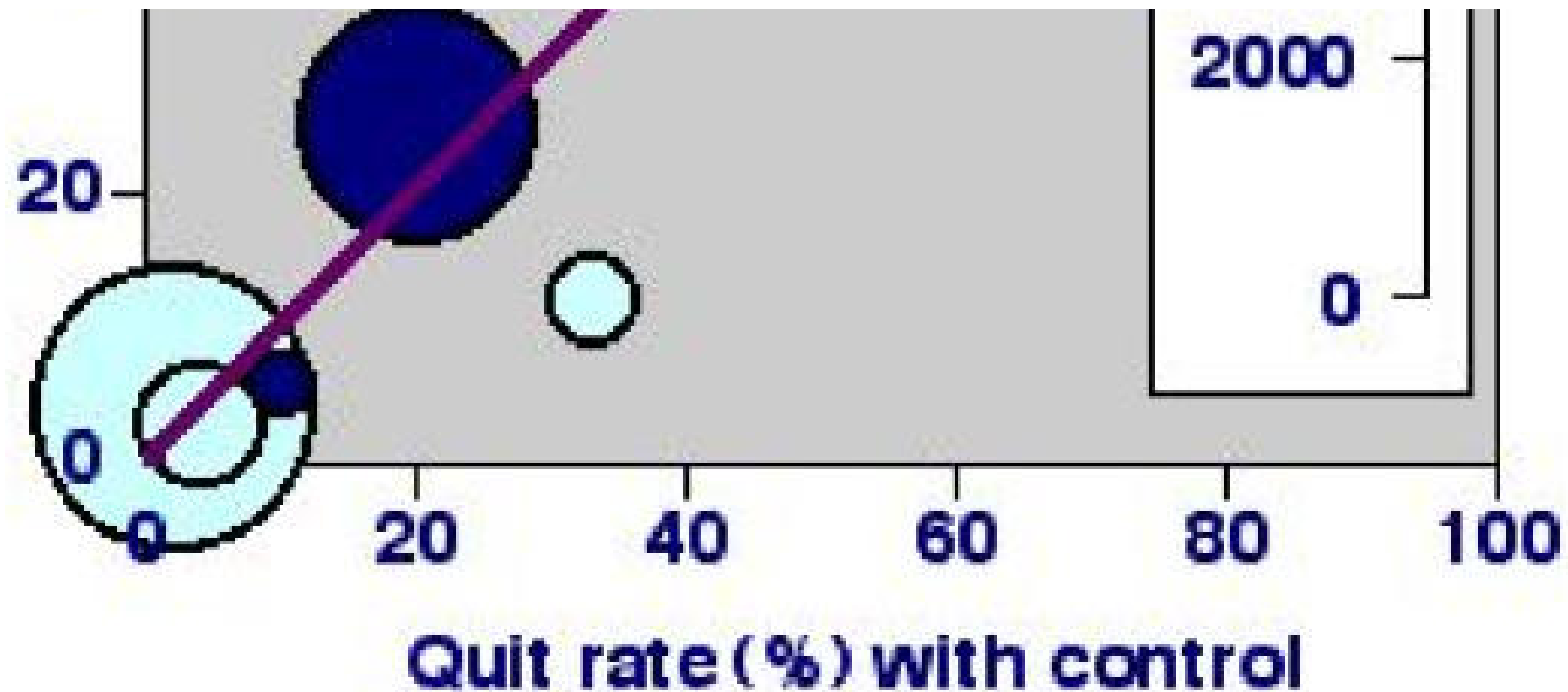
Not answered

Marked out of 1

The below graph is best described as

Quit rate (%) with intervention





Select one:

- ☐ Funnel plot
- ☐ Forest plot
- ☐ Galbraith plot
- ☐ Altman plot
- ☐ L'Abbe plot

Your answer is incorrect.

This is the L'Abbe plot which graphs the event rate in the treated group against the event rate in the control group. Note that L'Abbe plots using raw data will almost always show up heterogeneity, whereas Forest plots, in which data have been manipulated to create statistical outputs like odds ratios or risk differences, may or may not show heterogeneity.<http://www.biomedcentral.com/1471-2288/2/2/figure/F1>

The correct answer is: L'Abbe plot

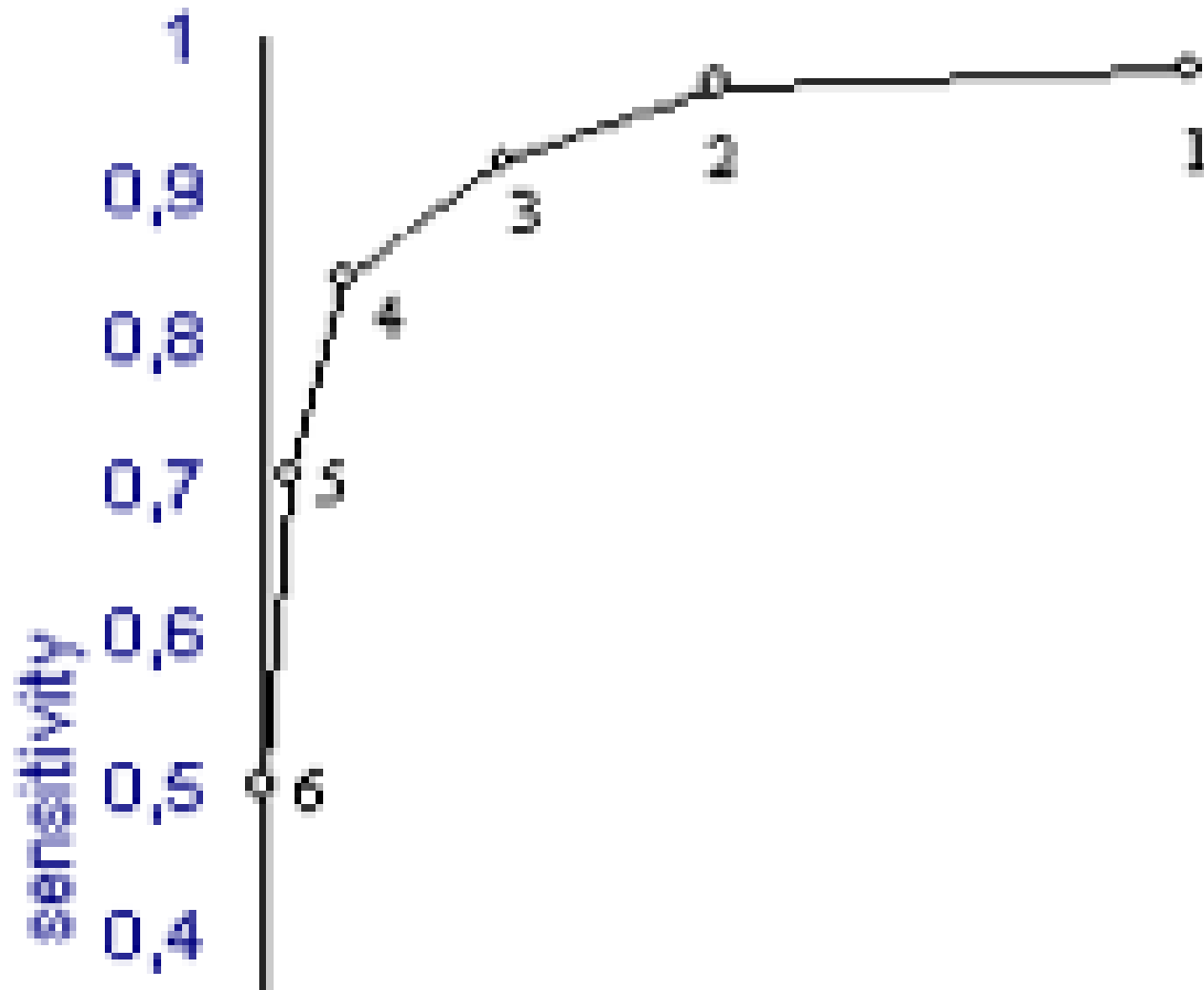


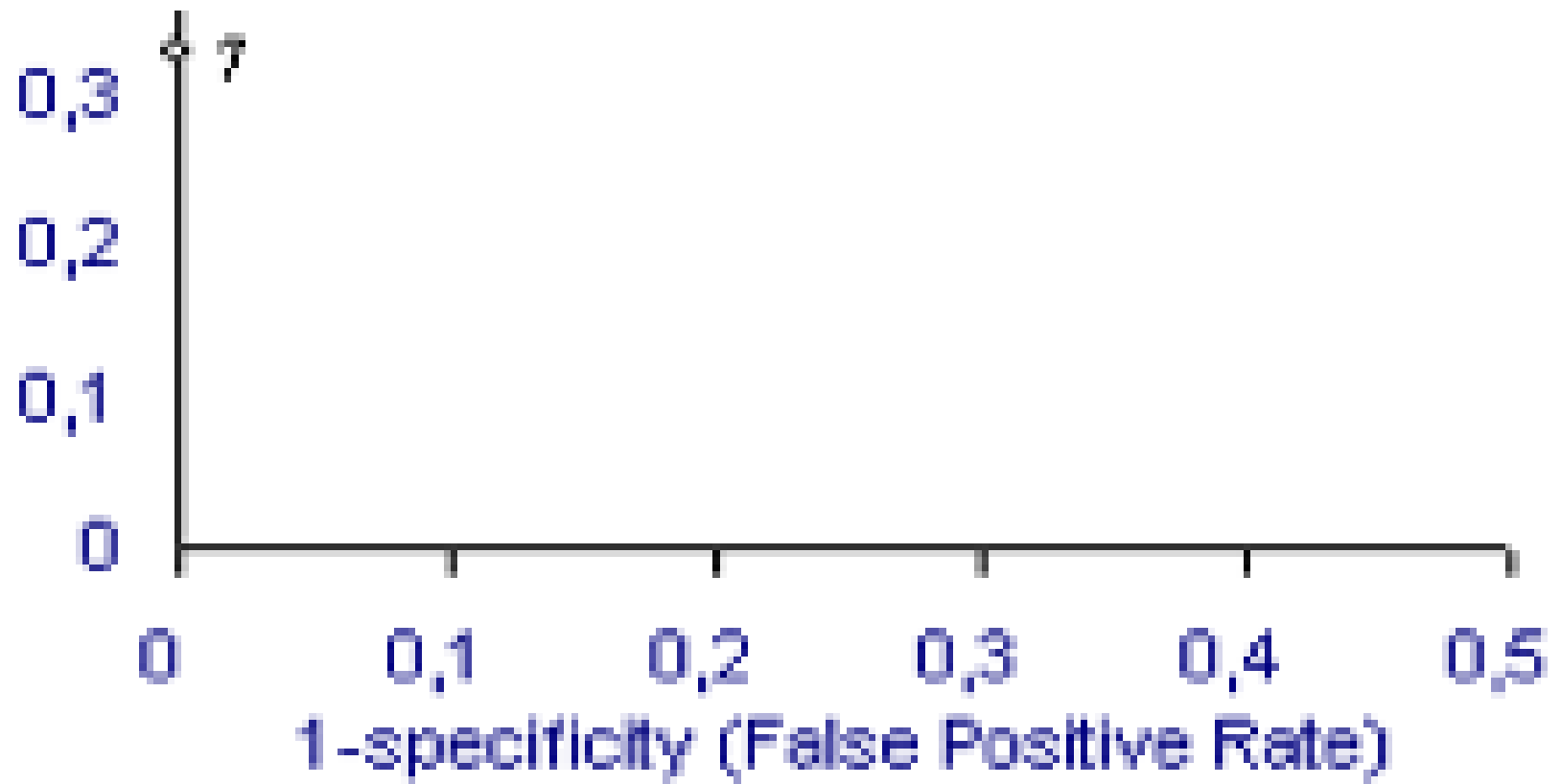
Question 6

Not answered

Marked out of 1

The below curve represents





Select one:

- ☐ Scatter gram
- ☐ Stem-leaf plot
- ☐ Funnel plot
- ☐ Galbraith plot
- ☐ Receiver operator curve

Your answer is incorrect.

Receiver operator curve plots sensitivity against false positive rate. It is useful when 1. Choosing between two diagnostic tests of different sensitivity and specificity rates 2. Choosing a cut-off point for making a diagnosis.

The correct answer is: Receiver operator curve

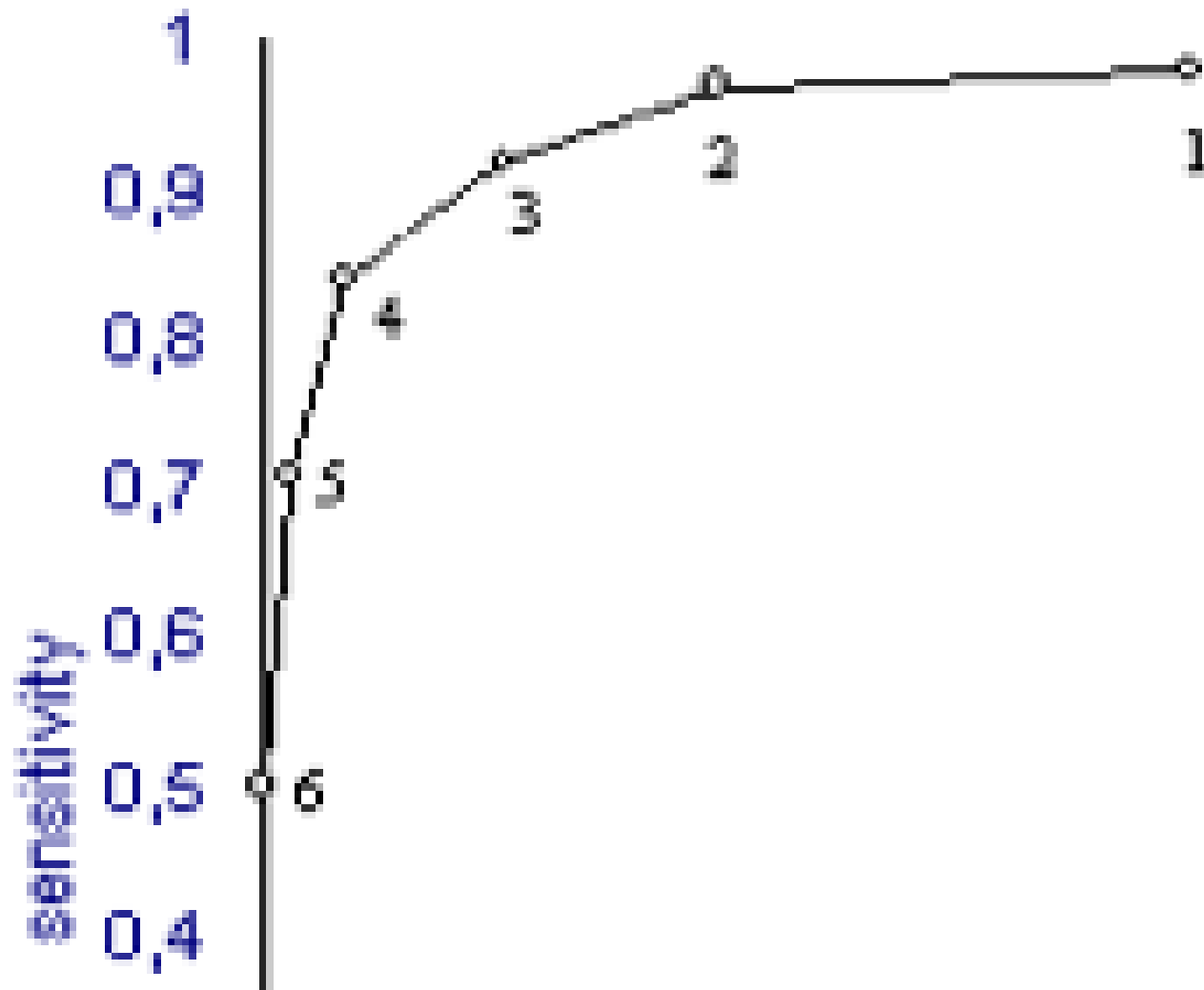


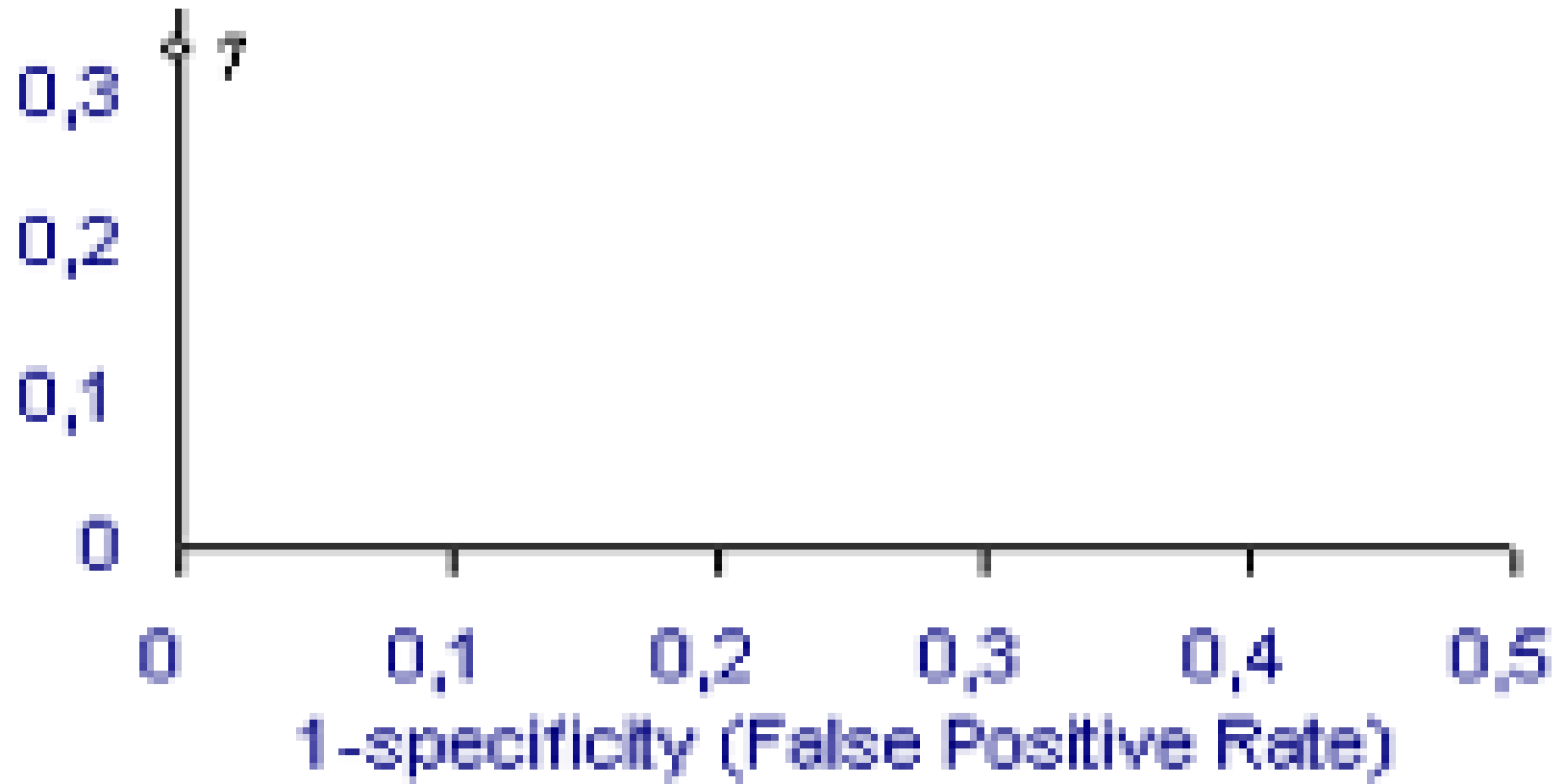
Question 7

Not answered

Marked out of 1

In the below curve (see the uploaded image) which number represents the point with the best trade-off between sensitivity and specificity?





Select one:

- ☐ 5
- ☐ 1
- ☐ 4
- ☐ 3
- ☐ 2

Your answer is incorrect.

By plotting the true positive rates (sensitivity) on the y axis and corresponding false positive rates (1-specificity) on the x-axis, the Receiver operator curve is obtained. The knee or elbow of the curve is a cut off point that has a balance between sensitivity and specificity. Here it is point 4. Among two different test curves, the one closer to upper left corner is the better screening test.

The correct answer is: 4

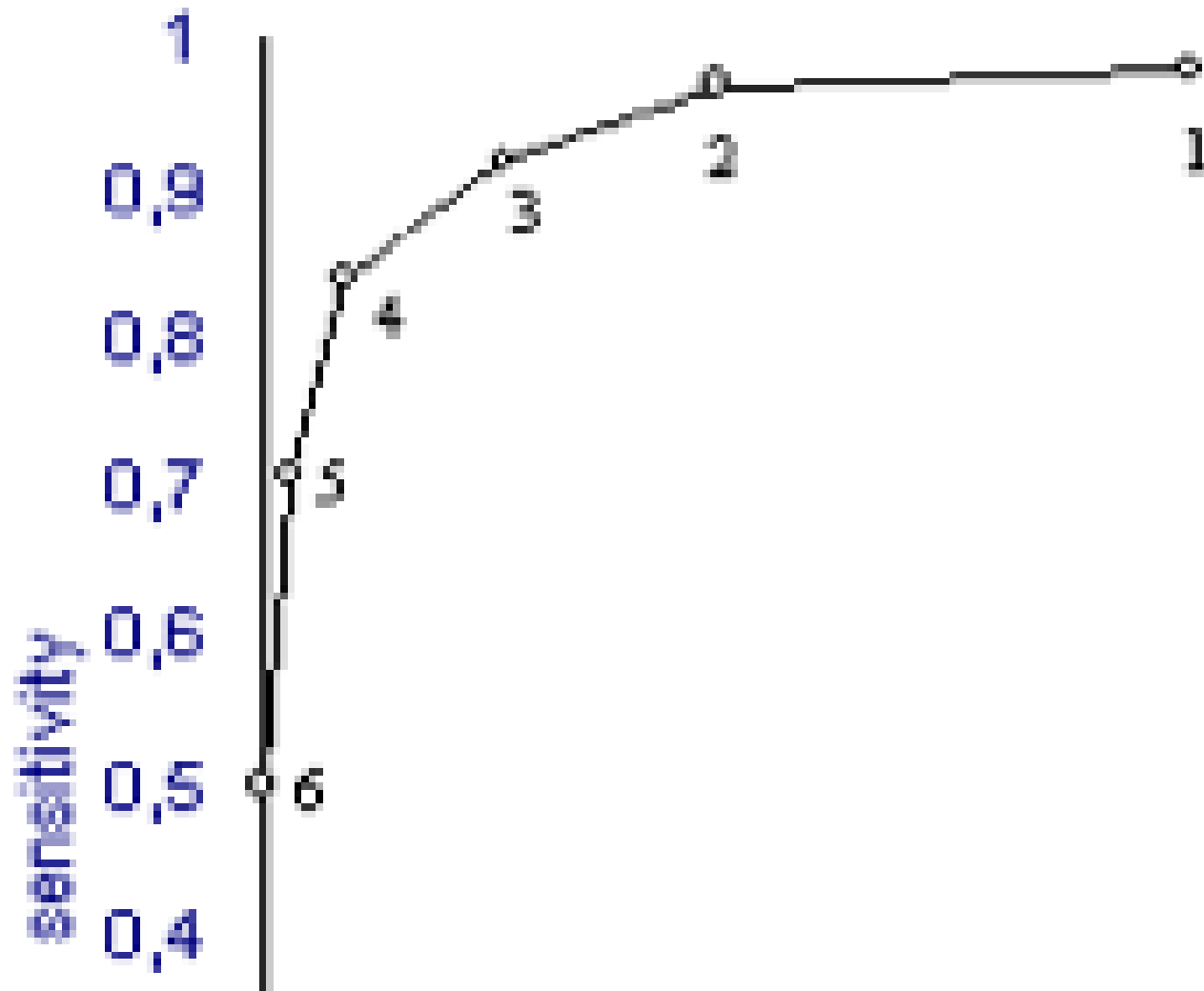


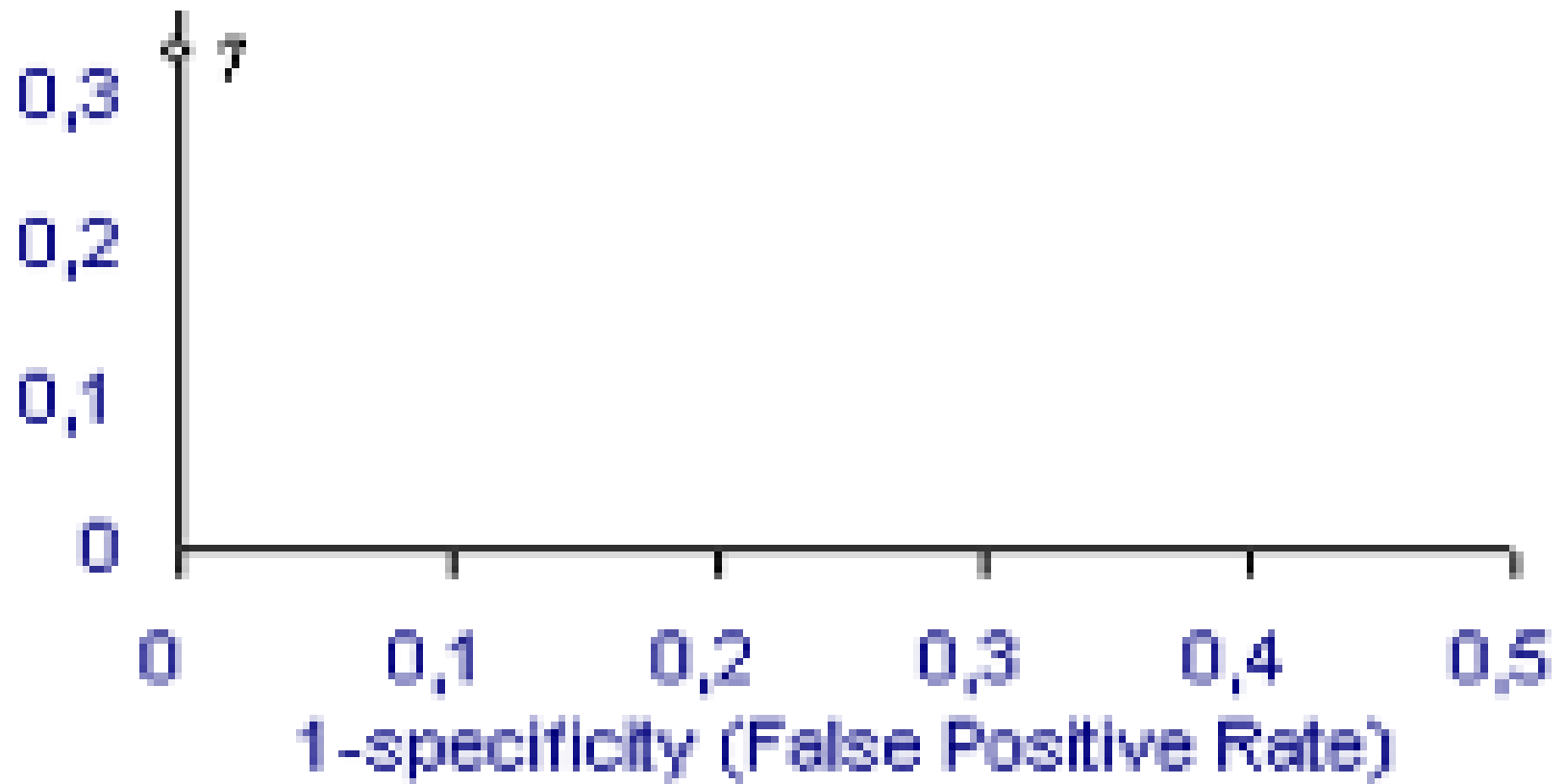
Question 8

Not answered

Marked out of 1

In the below graph, the area under the curve is representative of





Select one:

- ☐ Positive Predictive Value
- ☐ 1-specificity
- ☐ Sensitivity
- ☐ Accuracy
- ☐ 1-sensitivity

Your answer is incorrect.

ROC analysis is used to select the optimal cut point to dichotomize a continuous scale. The usual choice of cut points minimizes the overall number of false positive and false negative errors. The cut point may be shifted if the cost of false positives is higher than that of false negatives or vice versa. The accuracy of ROC analysis depends on the quality of the gold standard, which is usually far from perfect in psychiatry. Changing the purpose of the test (for example, from diagnosis to screening) requires a shift in cut points. A cut point that is ideal for one group may be less than ideal for another. It shows the trade-off between sensitivity and specificity (any increase in sensitivity will be accompanied by a decrease in specificity). The closer the curve follows the left-hand border and then the top border of the ROC space, the more accurate the test. The closer the curve comes to the 45-degree diagonal of the ROC space, the less accurate the test. The area under the curve is a measure of test accuracy. From <http://gim.unmc.edu/dxtests/ROC2.htm>

The correct answer is: Accuracy

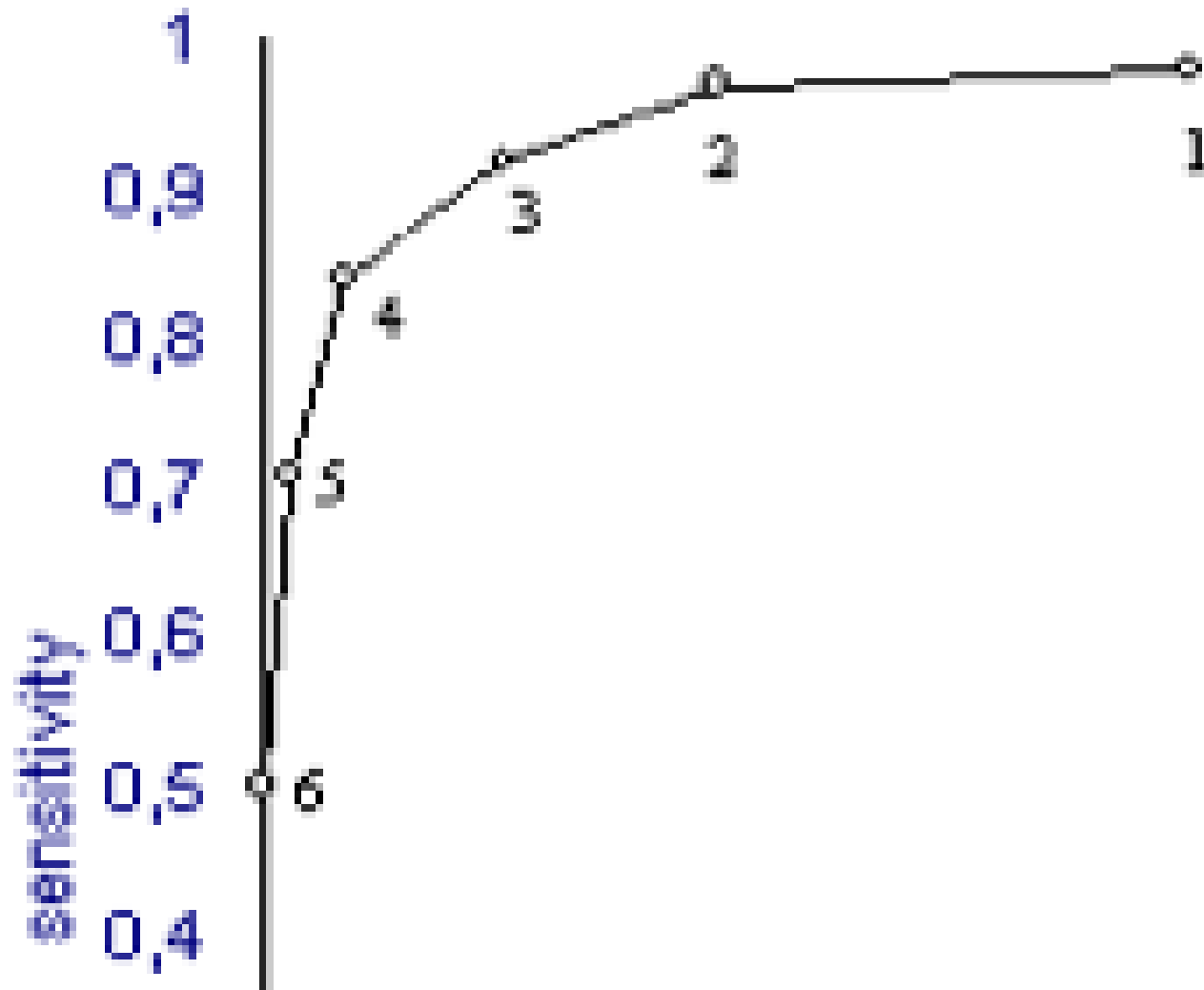


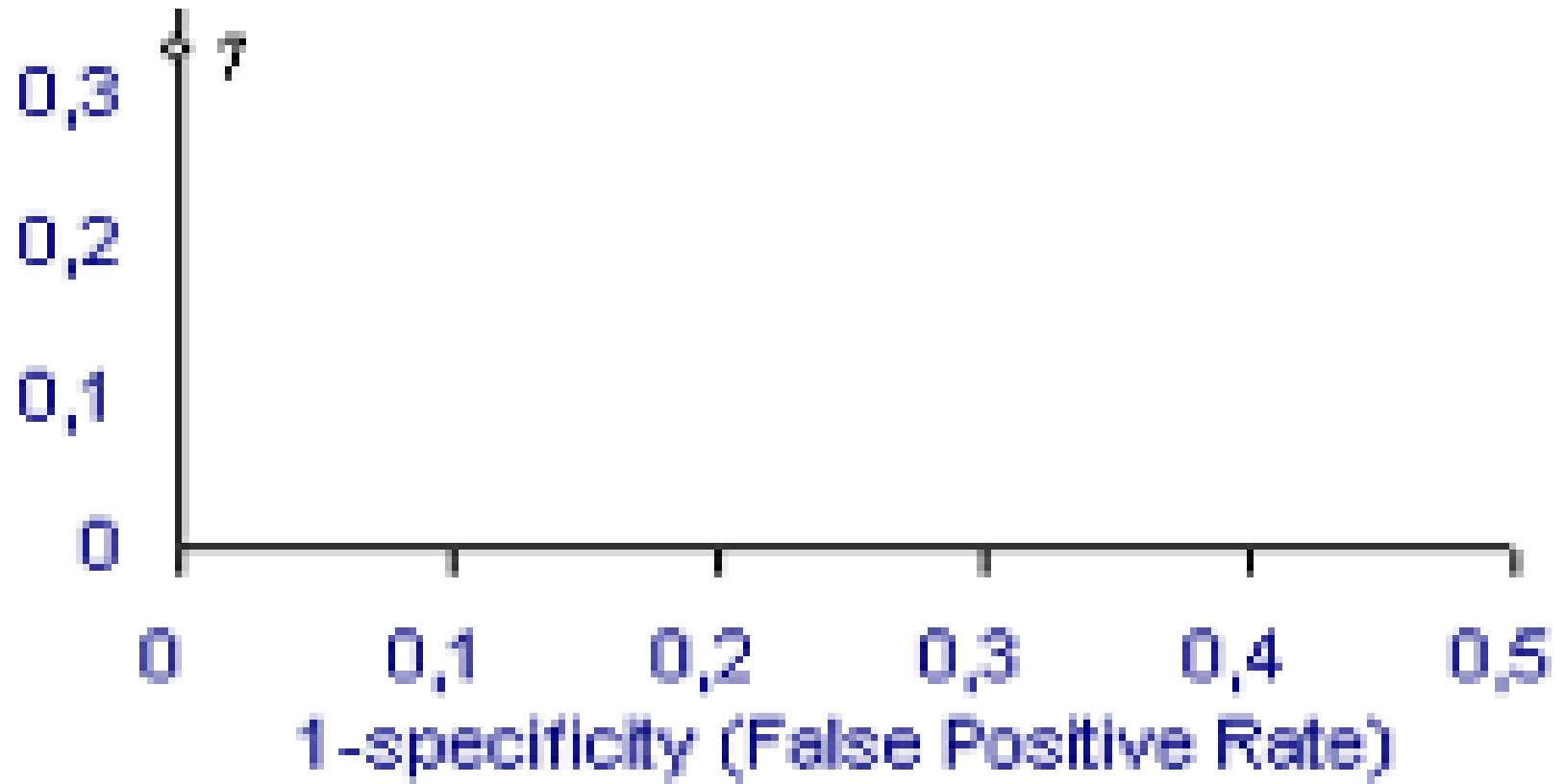
Question 9

Not answered

Marked out of 1

In the below graph, which value of cut-off would have the lowest specificity?





Select one:

- ☐ 4
- ☐ 6
- ☐ 1
- ☐ 2
- ☐ 3

Your answer is incorrect.

Please note that the x-axis in ROC represents 1-specificity, not just specificity. As the cutoff point 1 represents highest value for (1-specificity), it will be the point where the specificity is the lowest.

The correct answer is: 1



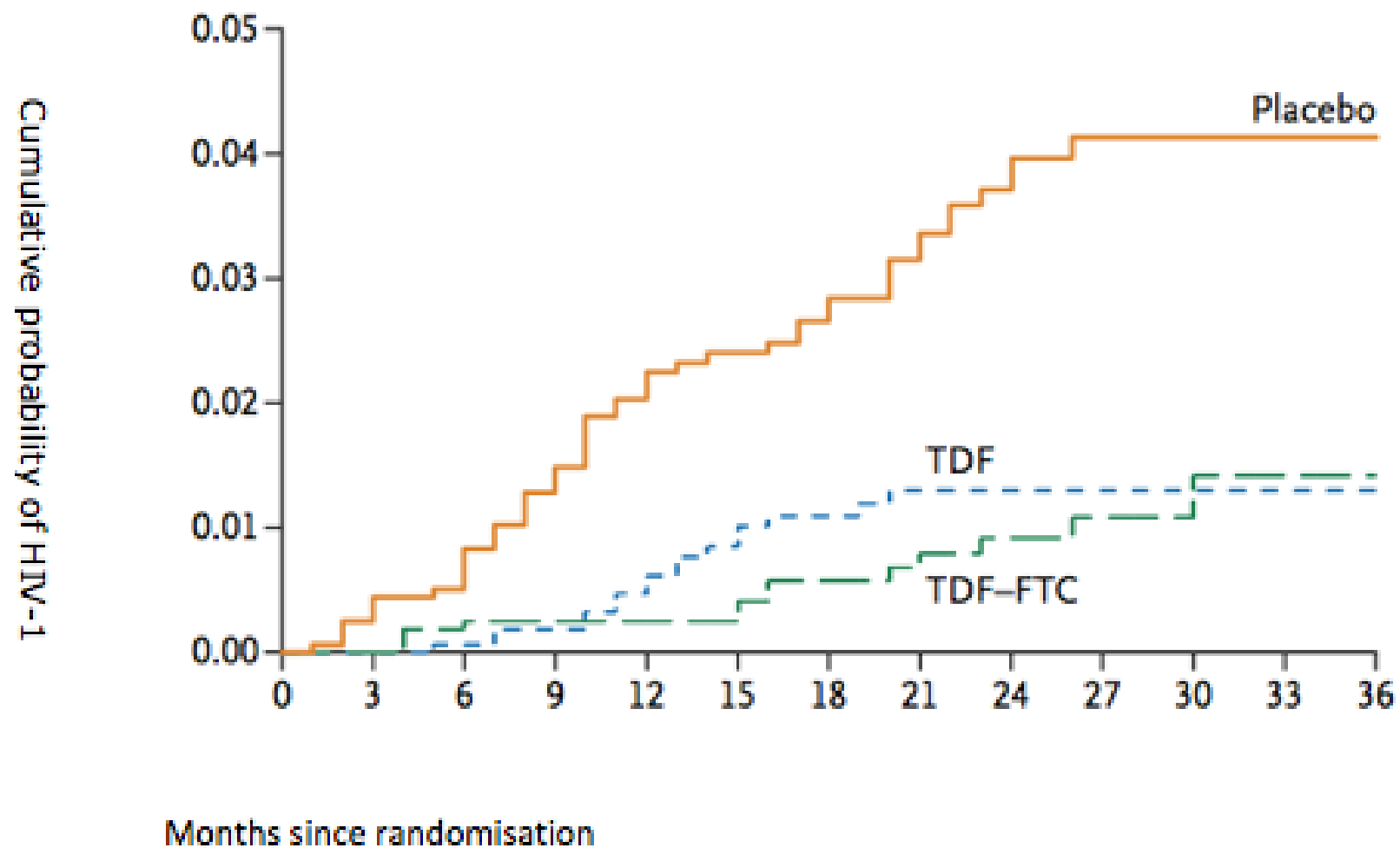
Question 10

Not answered

Marked out of 1

A randomized trial of oral antiretroviral therapy was conducted to evaluate the use of antiretroviral treatment as preexposure prophylaxis among HIV-1 –serodiscordant heterosexual couples from Kenya and Uganda. The HIV-1 –seronegative partner in each couple, was randomly assigned to one of the three study regimens — once-daily tenofovir (TDF), combination tenofovir – emtricitabine (TDF –FTC), or matching placebo — and followed monthly for up to 36 months. Of the enrolled 4758 couples 4747 were followed: 1584 randomly assigned to TDF, 1579 to TDF –FTC, and 1584 to placebo. A total of 82 HIV-1 infections occurred in seronegative participants during the study, 17 in the TDF group, 13 in the TDF –FTC group and 52 in the placebo group, indicating a relative reduction of 67% in the incidence of HIV-1 with TDF (95% confidence interval [CI], 44 to 81; $P < 0.001$) and of 75% with TDF –FTC (95% CI, 55 to 87; $P < 0.001$). Protective effects of TDF –FTC and TDF alone against HIV-1 were not significantly different ($P = 0.23$), and both study medications significantly reduced the HIV-1 incidence among both men and women. Eight participants receiving active treatment were found to have been infected with HIV-1 at baseline, and among these, antiretroviral resistance developed in two during the study. An estimate of the primary endpoint in the modified Intention-to-Treat Analysis is shown in a graph below. Which of the following inferences can be made from the graph?





Select one:

- ☐ By the end of the follow-up all couples were seropositive in the placebo arm
- ☐ The protective effect of antiretroviral treatment does not begin immediately
- ☐ The dropouts were highest for placebo group

- ☐ TDF and TDF-FTC groups were contaminated by more subjects violating treatment protocol in these groups
- ☐ Probability of HIV-1 infection increases with time

Your answer is incorrect.

By the end of follow-up (36 mo) around 4-5% of couples were seropositive in the placebo arm. This rate was even lower (1-2%) in those who receive antiretroviral treatment. The crossing of survival lines around 30 to 36 months for TDF and TDF-FTC suggests that after 30 mo, the two groups have comparable survival rates. It does not mean that the groups are contaminated. It is impossible to comment on dropouts from these curves. While we cannot conclusively say that the protective effects of the treatment begins (or not) immediately, it is very clear that with time the rate of seroconversion increases.

The correct answer is: Probability of HIV-1 infection increases with time



Question 11

Not answered

Marked out of 1

Which of the following statements is inaccurate about the elements of a forest plot?

Select one:

- ☐ The midline of a forest plot represents a 'no effect' line
- ☐ Each study is represented by a block at the point estimate of intervention effect with a vertical line extending either side of the block
- ☐ The area of the block indicates the weight assigned to that study
- ☐ In some forest plots, proportionately sized dots or squares may be used to ascribe the weighting assigned to each study
- ☐ The solid diamond represents the overall OR estimate of the studies presented.

Your answer is incorrect.

Each study is represented by a block at the point estimate of intervention effect with a horizontal line extending either side of the block. The weighting is usually assigned based on the relative strengths such as sample size, methodology of the individual studies.

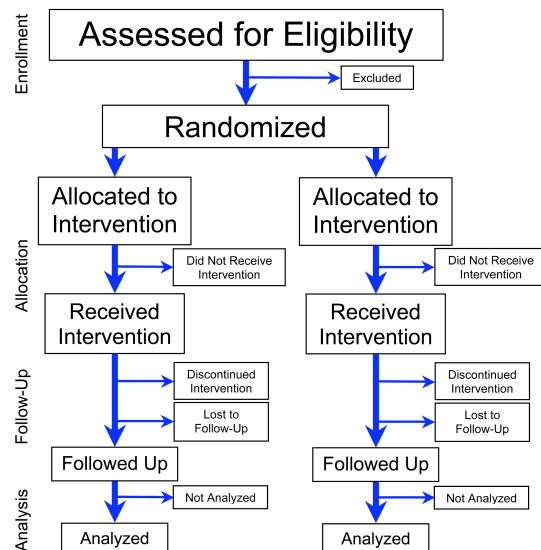
The correct answer is: Each study is represented by a block at the point estimate of intervention effect with a vertical line extending either side of the block

Question 12

Not answered

Marked out of 1

Which one of the following information is strongly recommended to be always included in this diagram? Please see the below graph.



Select one:

- ☐ Inconvenience allowance paid to subjects
- ☐ Ethnic origin of participants
- ☐ Attrition rate
- ☐ The costs incurred for the trial
- ☐ Integrity of blinding

Your answer is incorrect.

It is strongly recommended that the number of dropouts (attrition rate) is clearly shown in these diagrams.

The correct answer is: Attrition rate

Question **13**

Not answered

Marked out of 1

Which one of the following is a recommendation made by the CONSORT group for reporting RCTs?

Select one:

- ☐ Explicitly report of the costs incurred for the trial
- ☐ Include a participant flow diagram
- ☐ Preferably use parametric statistical tests
- ☐ Publish anonymised individual subject data in the public domain
- ☐ Publish trial results in open-access format

Your answer is incorrect.

A flow diagram is intended to depict the passage of participants through an RCT. Reporting of the flow of participants through each stage of a randomized trial is essential to assess the generalisability and validity of its results and is strongly recommended by the CONSORT statement.

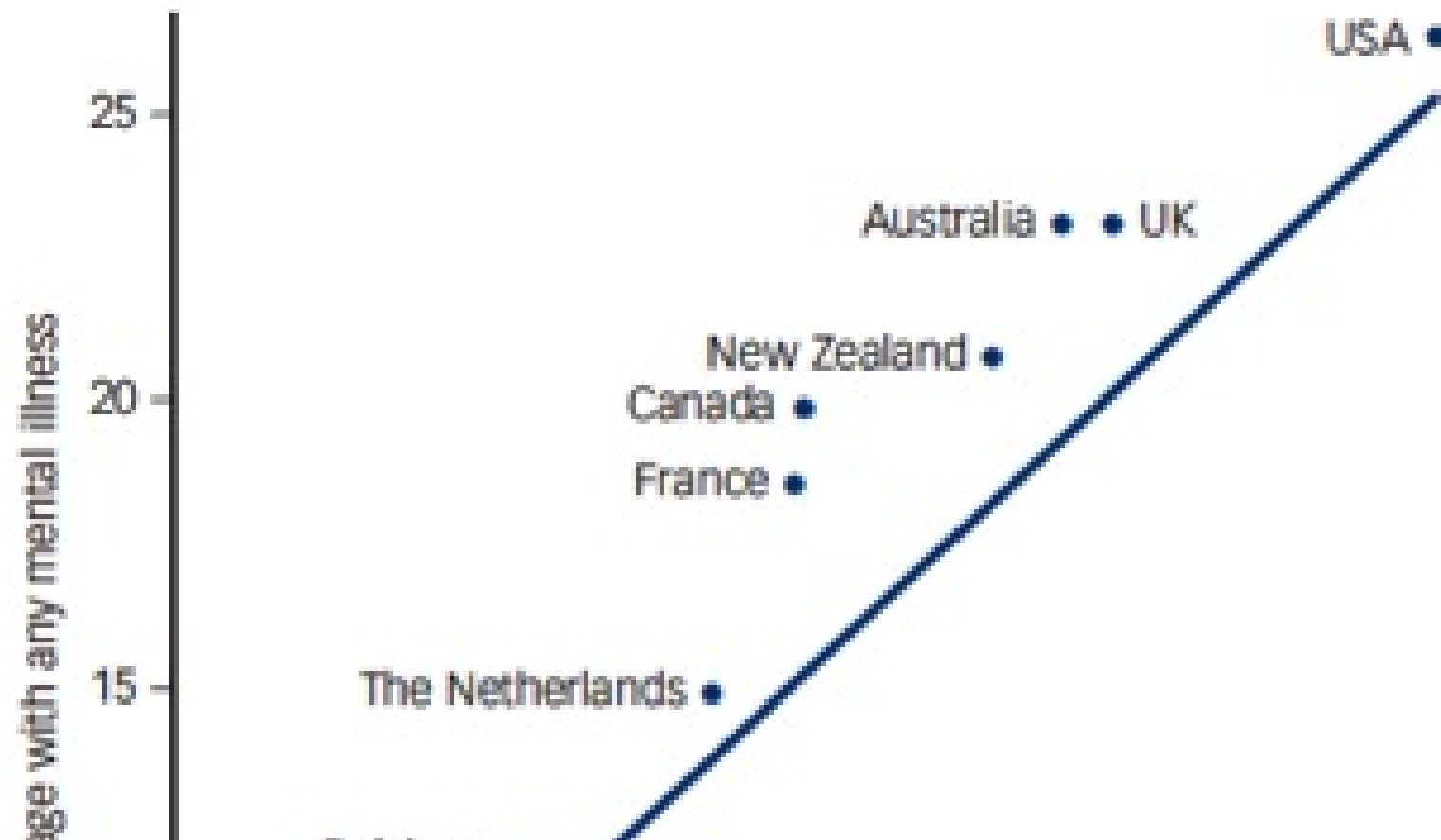
The correct answer is: Include a participant flow diagram

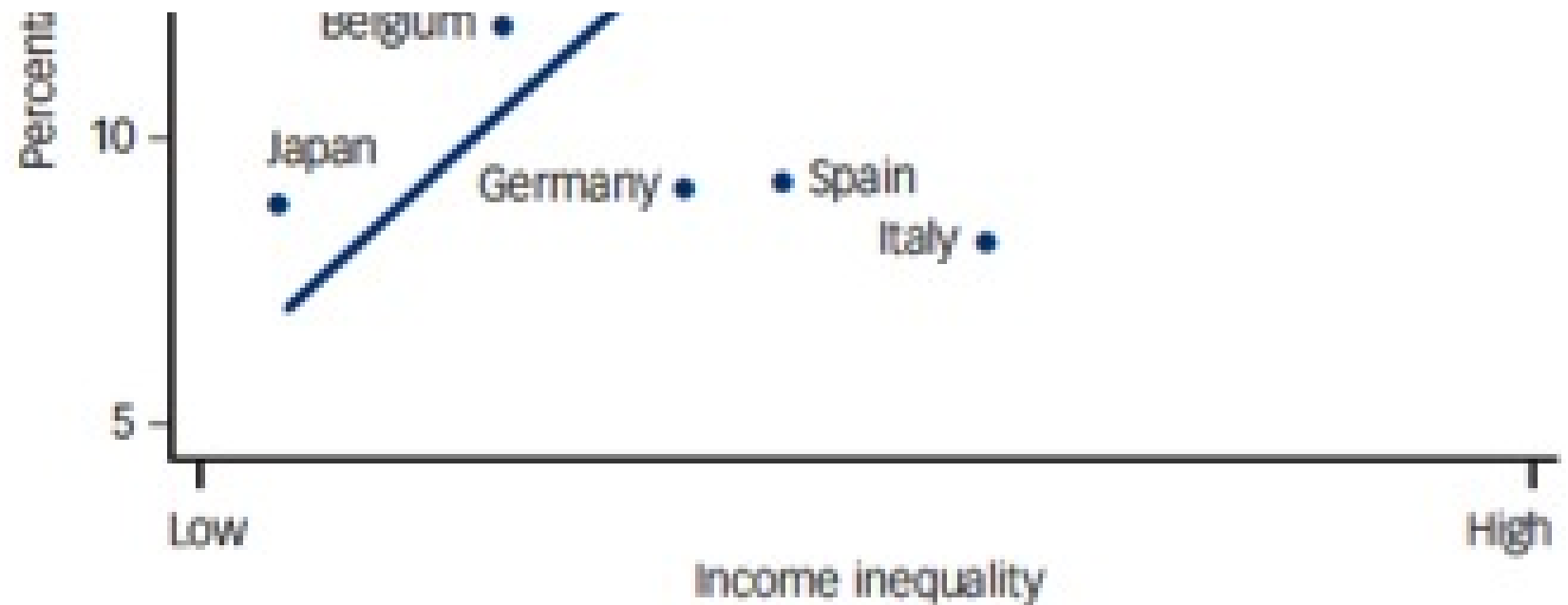
Question 14

Not answered

Marked out of 1

The data on the prevalence of mental illness in the World Health Organization surveys was used to investigate the impact of income equality on health. The below figure shows the association in 12 countries between income inequality and the proportion of adults who have been mentally ill in the 12 months prior to being interviewed. (Pickett KE & Wilkinson RG. (2010) Inequality: an underacknowledged source of mental illness and distress. The British Journal of Psychiatry 197,426-428). What can you say about the below graph? It is a





Select one:

- ☐ Cost effectiveness curve
- ☐ Cost benefit plot
- ☐ Prevalence chart
- ☐ Population pyramid
- ☐ Scatterplot

Your answer is incorrect.

The attached graph is a scatter plot. The dependent variable plotted on the y axis is the prevalence of mental illness in various countries. The independent variable on the x axis is income inequality in these countries.

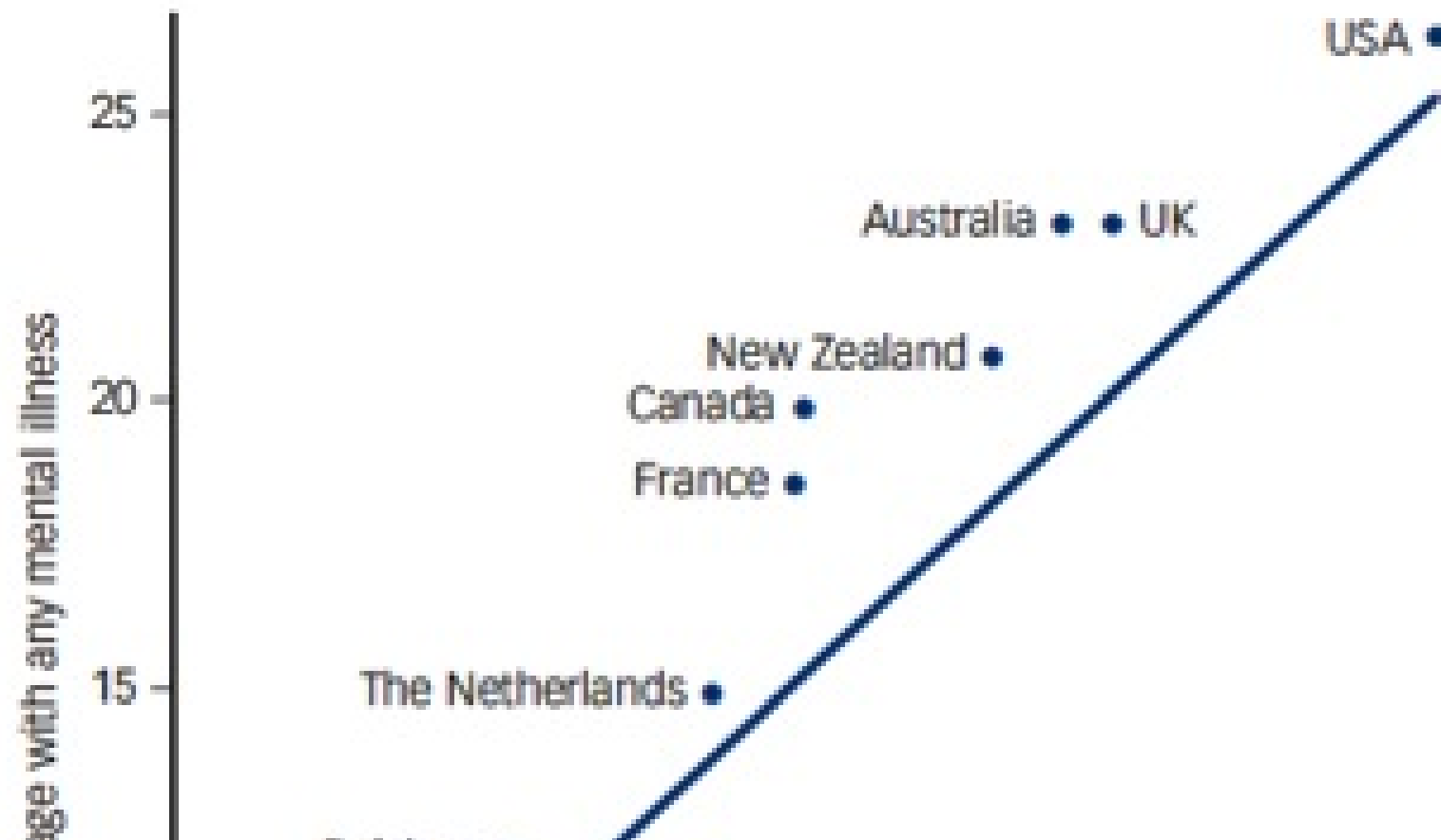
The correct answer is: Scatterplot

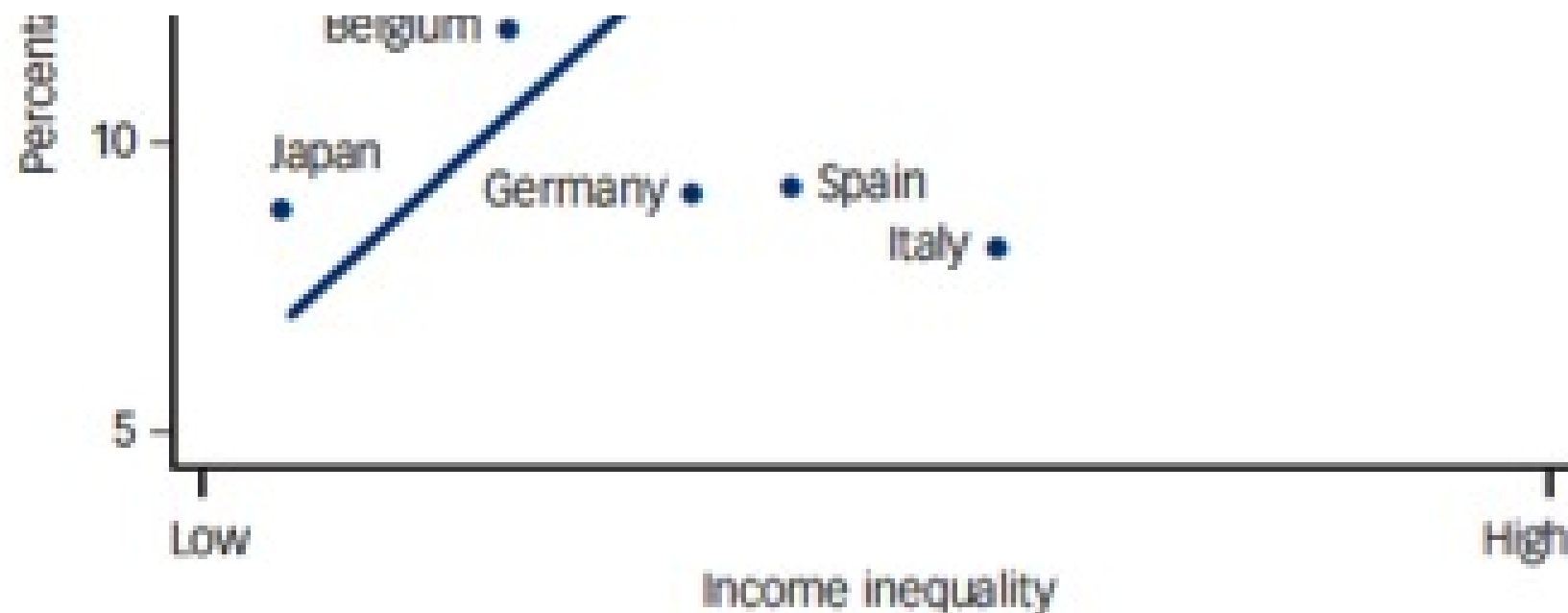
Question 15

Not answered

Marked out of 1

The data on the prevalence of mental illness in the World Health Organization surveys was used to investigate the impact of income equality on health. The below figure shows the association in 12 countries between income inequality and the proportion of adults who have been mentally ill in the 12 months prior to being interviewed. (Pickett KE & Wilkinson RG. (2010) Inequality: an underacknowledged source of mental illness and distress. The British Journal of Psychiatry 197,426-428). What can you conclude from the below graph?





Select one:

- ☐ Poverty is directly related to global rates of mental illness
- ☐ Countries with higher income inequality have higher rates of mental illness
- ☐ Higher rates of mental illness across different countries cause income inequality
- ☐ Income inequality causes mental illness across different countries
- ☐ Countries with higher income rates have higher rates of mental illness

Your answer is incorrect.

The linear line fitting the data on the scatter plot has a positive slope suggesting a direct positive relationship between income inequality and the prevalence of mental illness in the countries studied here.

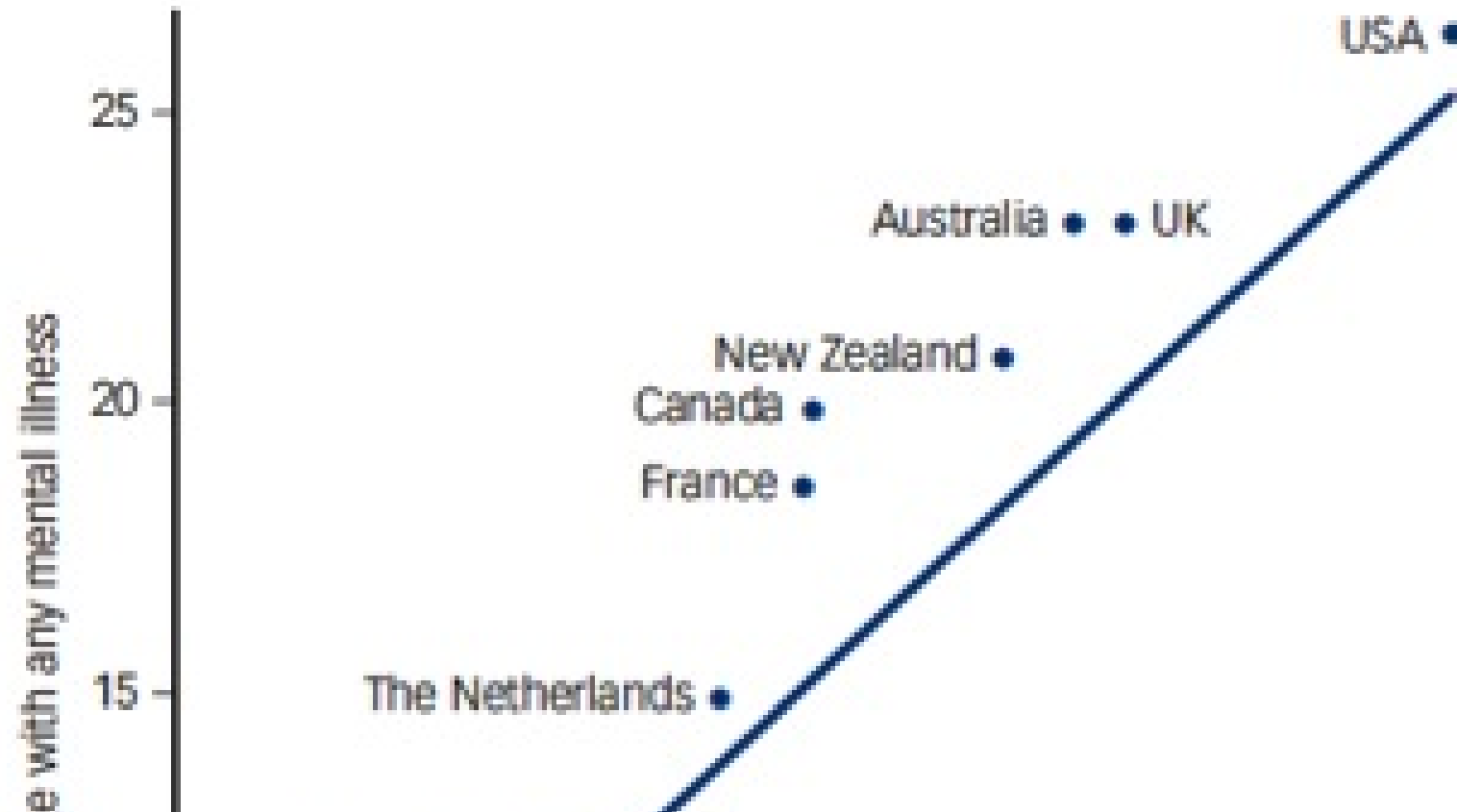
The correct answer is: Countries with higher income inequality have higher rates of mental illness

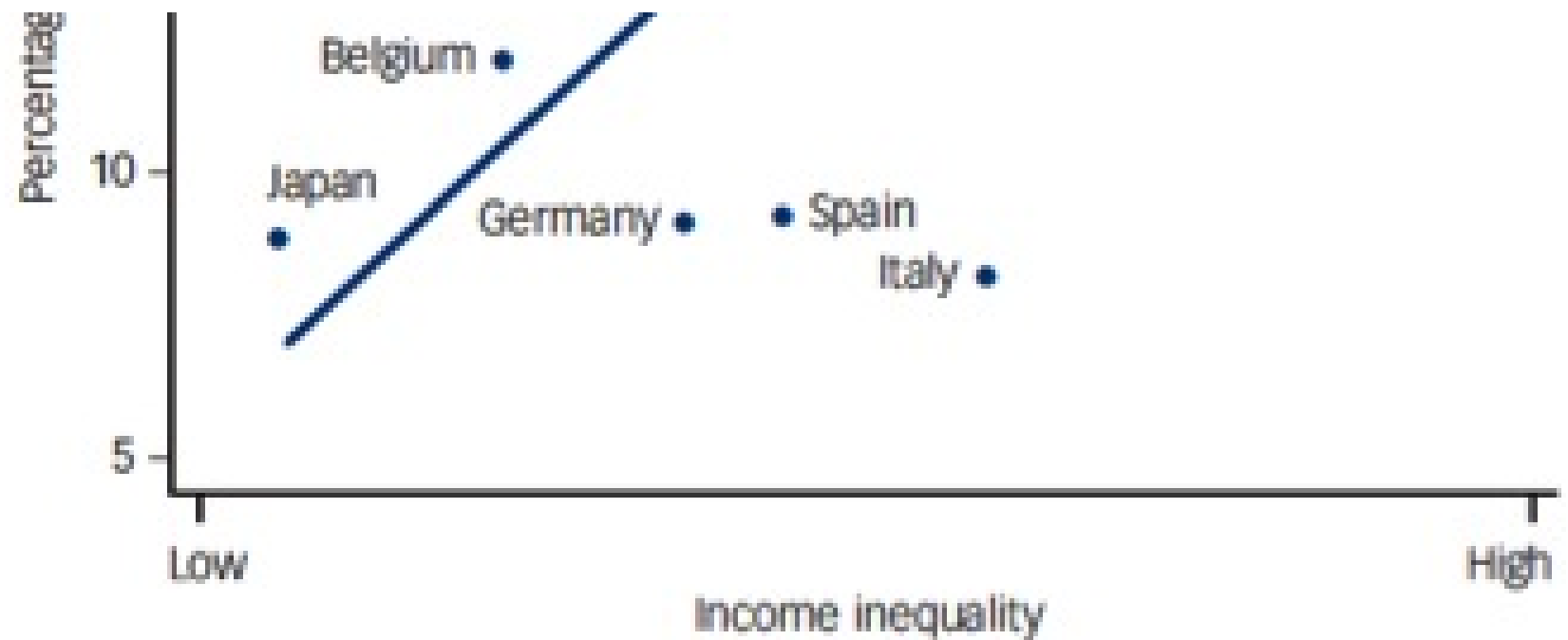
Question 16

Not answered

Marked out of 1

The data on the prevalence of mental illness in the World Health Organization surveys was used to investigate the impact of income equality on health. The below figure shows the association in 12 countries between income inequality and the proportion of adults who have been mentally ill in the 12 months prior to being interviewed. (Pickett KE & Wilkinson RG. (2010) Inequality: an underacknowledged source of mental illness and distress. The British Journal of Psychiatry 197,426-428). Which one of the following countries can be identified as an outlier in the below graph?





Select one:

- ☐ USA
- ☐ Spain
- ☐ Japan
- ☐ Italy
- ☐ Germany

Your answer is incorrect.

Italy does not show the relationship between income inequality and proportion of mentally ill population, unlike other countries. The linear model fitted to the data shows that Italy is far away or deviant from the line.

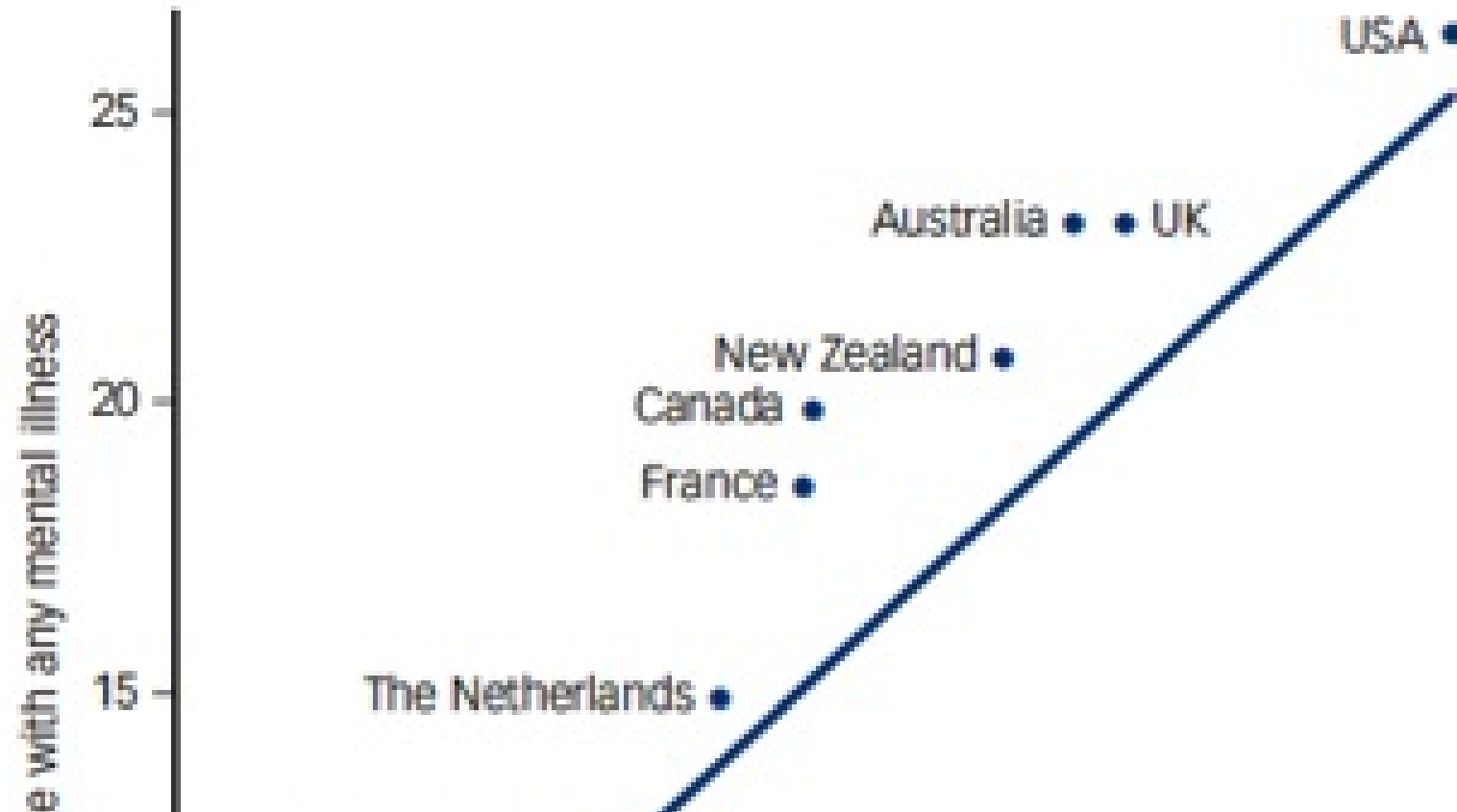
The correct answer is: Italy

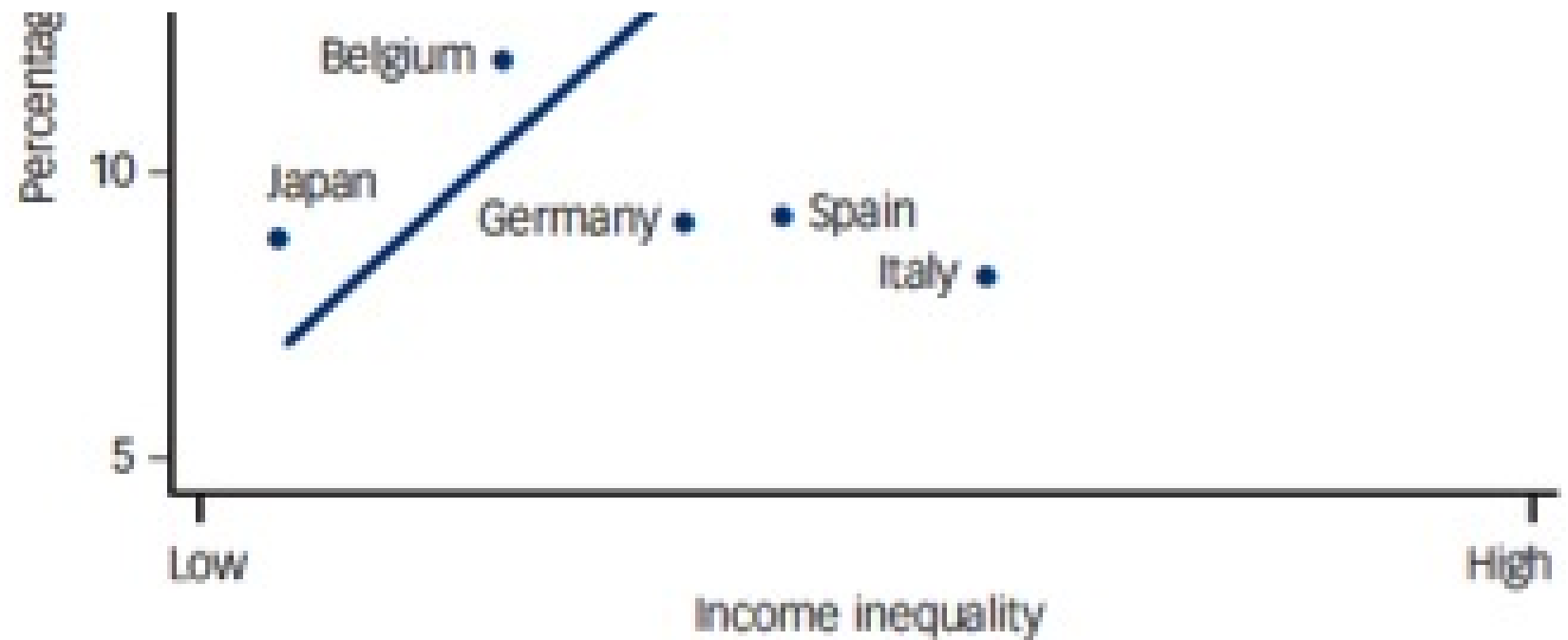
Question 17

Not answered

Marked out of 1

The data on the prevalence of mental illness in the World Health Organization surveys was used to investigate the impact of income equality on health. The below figure shows the association in 12 countries between income inequality and the proportion of adults who have been mentally ill in the 12 months prior to being interviewed. (Pickett KE & Wilkinson RG. (2010) Inequality: an underacknowledged source of mental illness and distress. The British Journal of Psychiatry 197,426-428). Which one of the following types of errors is most likely to influence the interpretation of the below study?





Select one:

- ☐ Attrition bias
- ☐ Ecological fallacy
- ☐ Recall bias
- ☐ Incidence-Prevalence bias
- ☐ Publication bias

Your answer is incorrect.

This is an ecological study (income inequality is studied in geographical clusters i.e. countries) and not at individual level. This type of studies suffer from ecological fallacy.

The correct answer is: Ecological fallacy

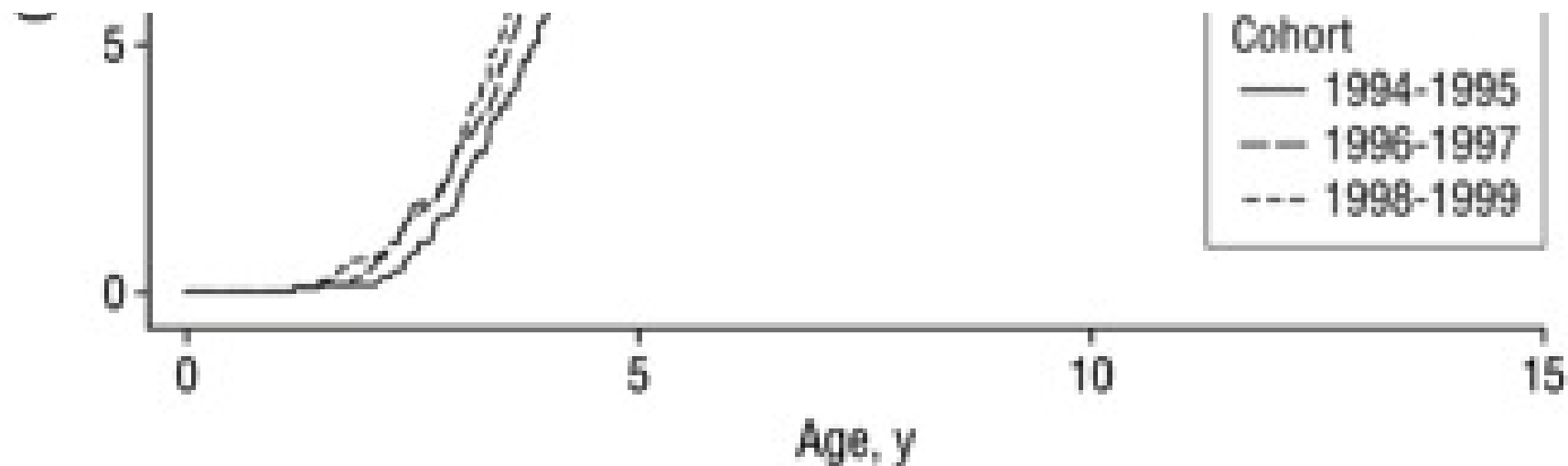
Question 18

Not answered

Marked out of 1

A study was conducted to examine trend-over-time in childhood autism using survival analysis methods [Arch Pediatr Adolesc Med. 2007;161(2):193-198.]. The below graph presenting the cumulative incidence proportion (per 10 000) for each 2-year analytic birth cohort can be termed as





Select one:

- ☐ Proportional incidence by age for autism across birth years
- ☐ Variation in the incidence of autism across three decades
- ☐ Time taken for survival once autism is diagnosed
- ☐ Variation in the incidence of autism in different countries
- ☐ Time delay in the diagnosis of autism

Your answer is incorrect.

"In 1958, Edward L. Kaplan and Paul Meier collaborated to publish a seminal paper on how to deal with incomplete observations. Subsequently, the Kaplan-Meier curves and estimates of survival data have become a familiar way of dealing with differing survival times (times-to-event), especially when not all the subjects continue in the study. "Survival" times need not relate to actual survival with death being the event; the "event" may be any event of interest". <http://www.ncbi.nlm.nih.gov/pubmed/20723767>

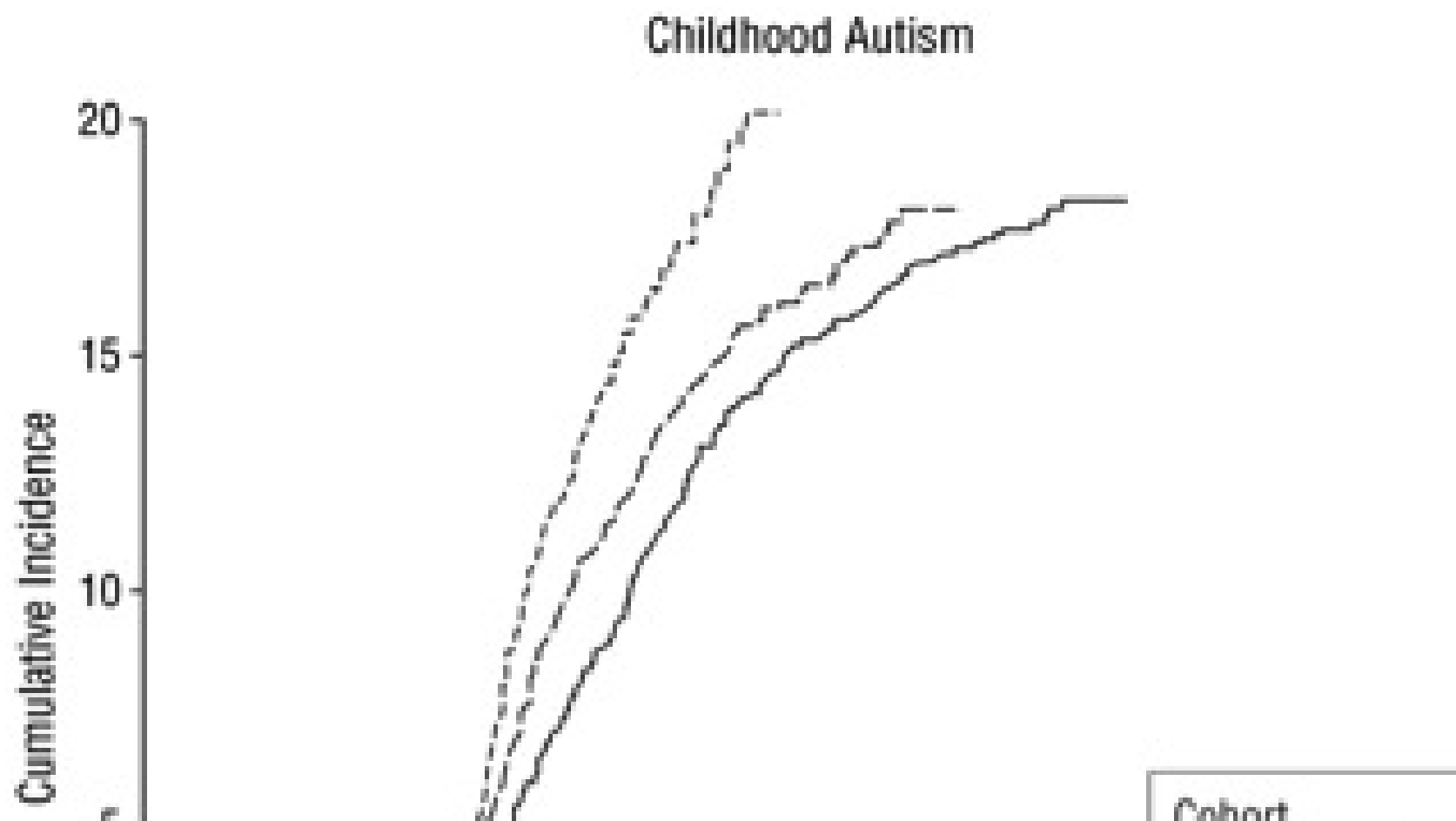
The correct answer is: Proportional incidence by age for autism across birth years

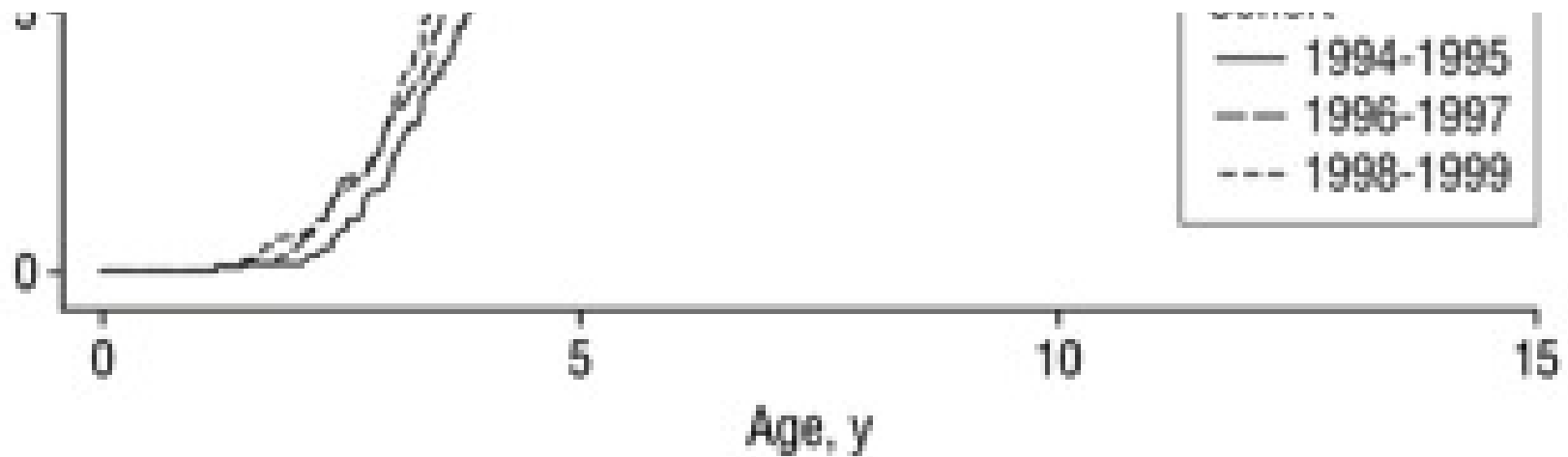
Question 19

Not answered

Marked out of 1

A study was conducted to examine trend-over-time in childhood autism using survival analysis methods [Arch Pediatr Adolesc Med. 2007;161(2):193-198.]. Which of the following is the primary outcome of interest in this study?





Select one:

- ☐ Incidence curve
- ☐ Scatter plot
- ☐ Linear log analysis
- ☐ ROC curve
- ☐ Kaplan Meier curve

Your answer is incorrect.

Main Outcome Measure in this study is cumulative incidence proportion by age, stratified by year of birth.

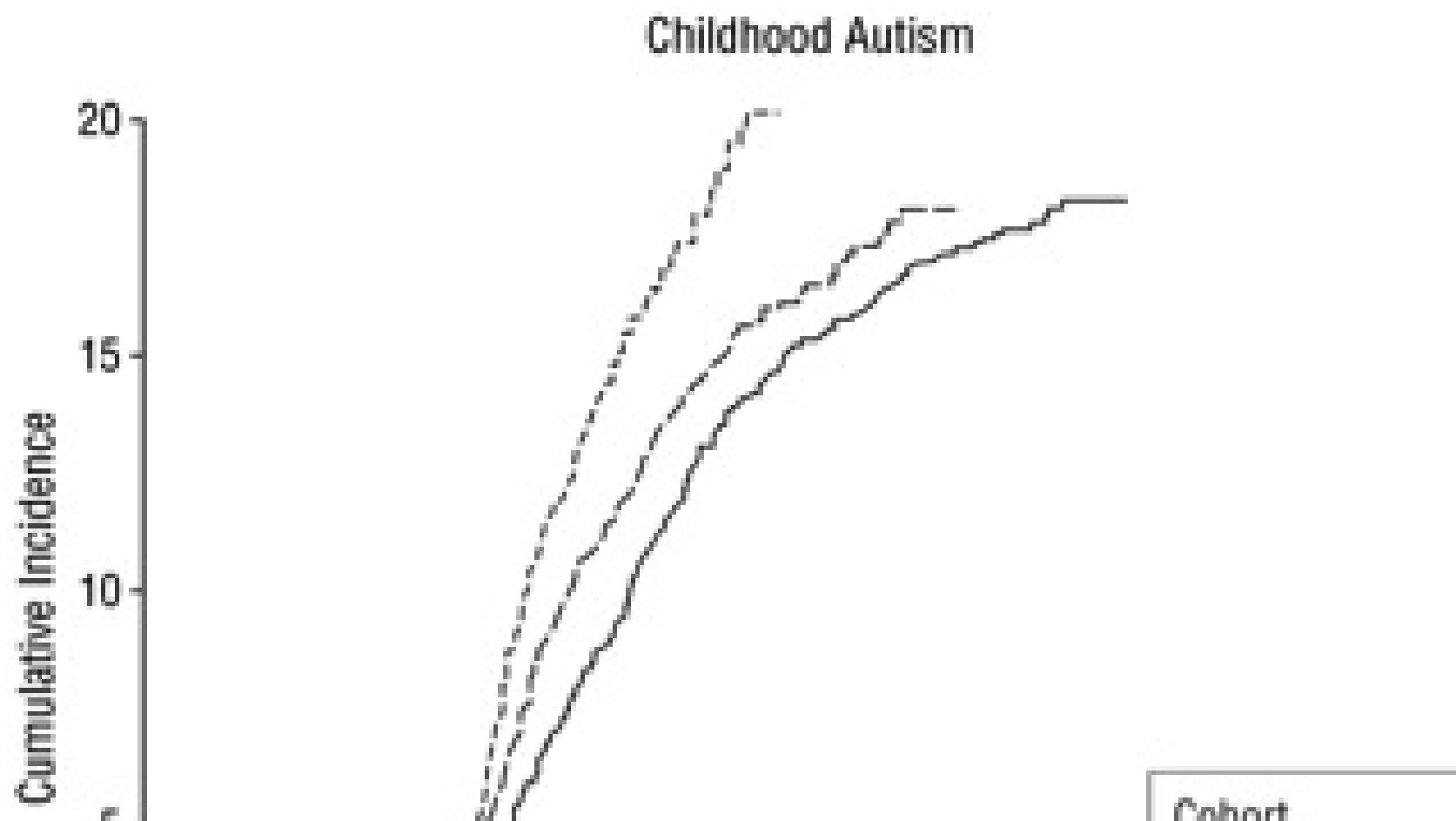
The correct answer is: Kaplan Meier curve

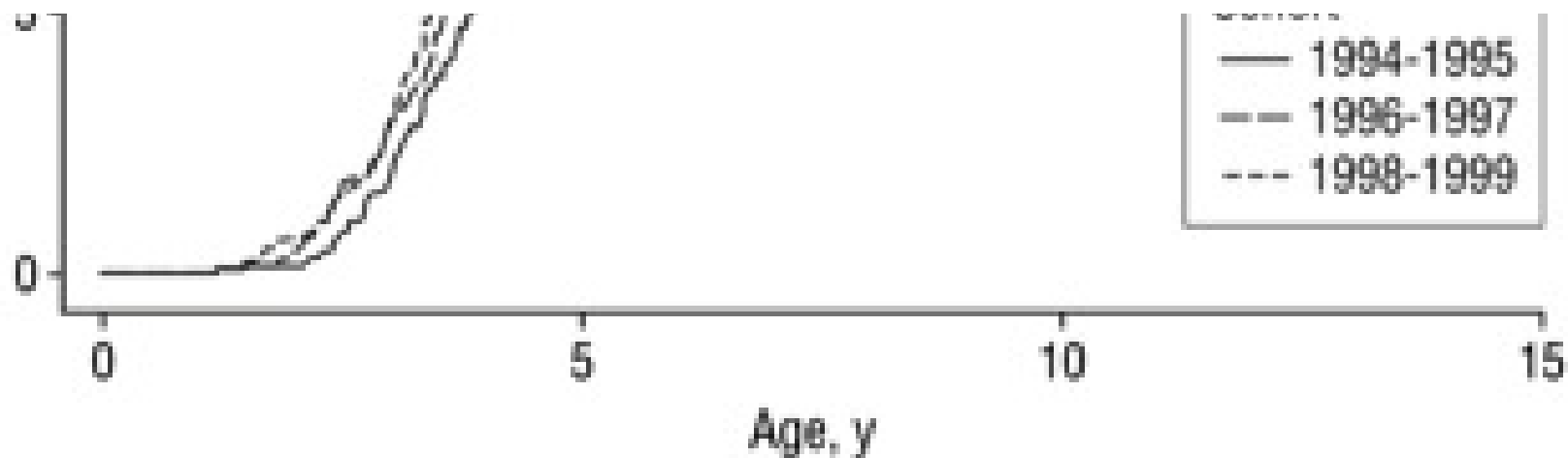
Question 20

Not answered

Marked out of 1

A study was conducted to examine trend-over-time in childhood autism using survival analysis methods [Arch Pediatr Adolesc Med. 2007;161(2):193-198.]. Which of the following study design is most likely to have been used in order to obtain the graph below?





Select one:

- ☐ Population-based cohort study
- ☐ Case control study
- ☐ Ecological study
- ☐ Cross-sectional study
- ☐ Case series

Your answer is incorrect.

Kaplan-Meier estimate is one of the best options to be used to measure the number of subjects survived (usually after an intervention) over a period of time. The time starting from a defined point to the occurrence of a given event (for example, death) is known as survival time and the analysis of group data as survival analysis. Hence prospective cohort designs (and RCTs) are best suited to generate this curve.

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3059453/>

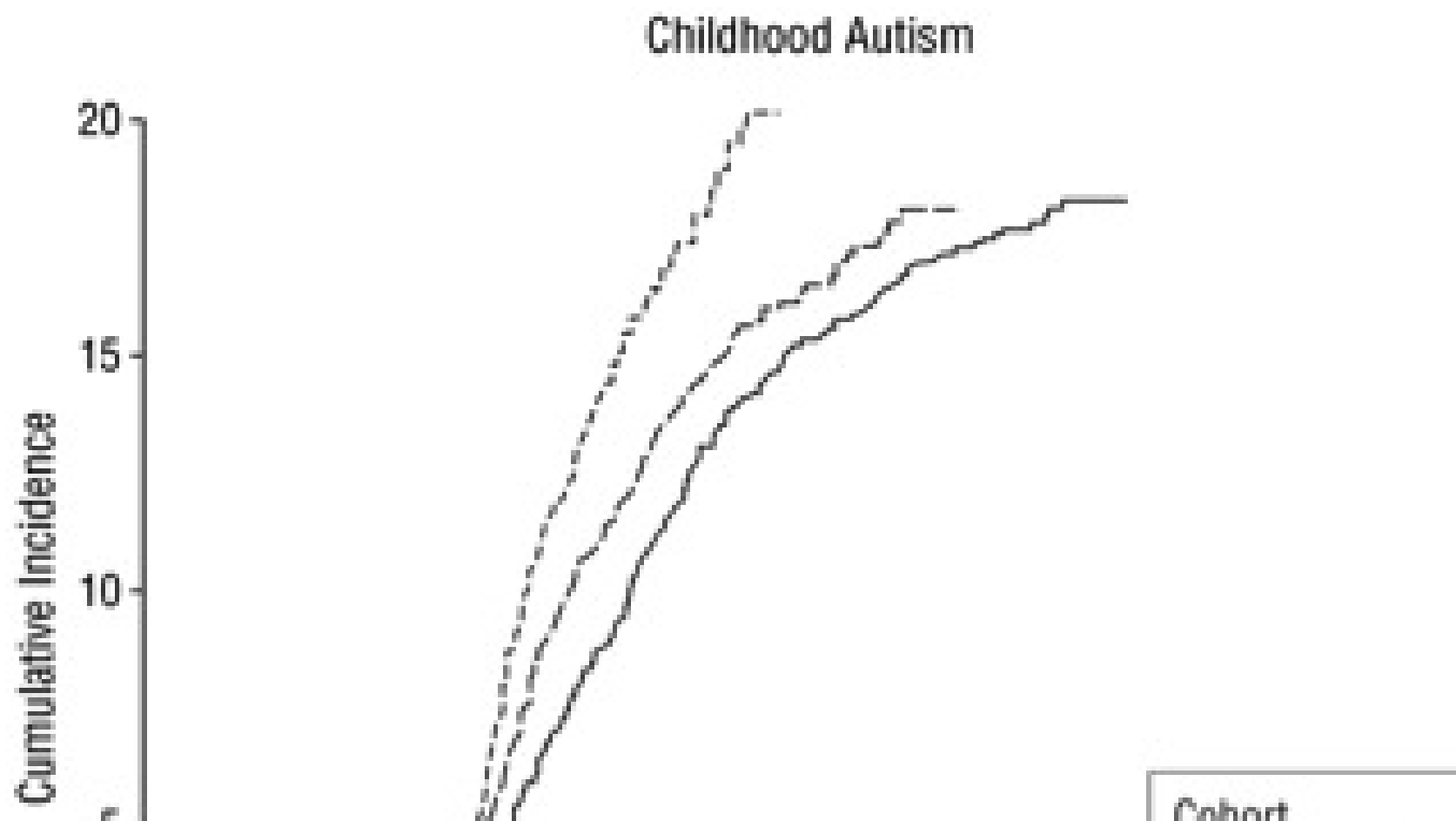
The correct answer is: Population-based cohort study

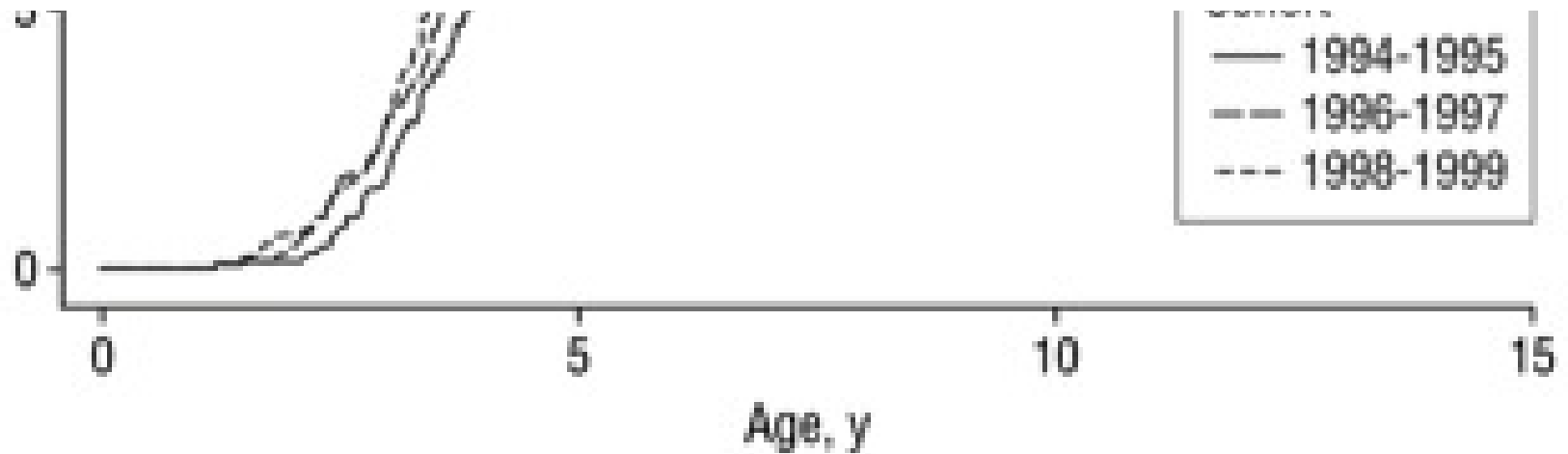
Question 21

Not answered

Marked out of 1

A study was conducted to examine trend-over-time in childhood autism using survival analysis methods [Arch Pediatr Adolesc Med. 2007;161(2):193-198.]. The advantage of the below graphical representation is that it demonstrates





Select one:

- ☐ The time taken for 100% of the population to achieve the event of interest
- ☐ The time delay for a clinical diagnosis of autism in each age group
- ☐ The effect of both age and year of birth on the cumulative incidence
- ☐ The age of peak incidence
- ☐ The effect of diagnostic practices on the incidence rates

Your answer is incorrect.

The cumulative incidence proportion for each 2-year birth cohort was estimated by using the Kaplan-Meier curve. Presenting data in this way demonstrates how both age and year of birth affect the cumulative incidence of a diagnosis.

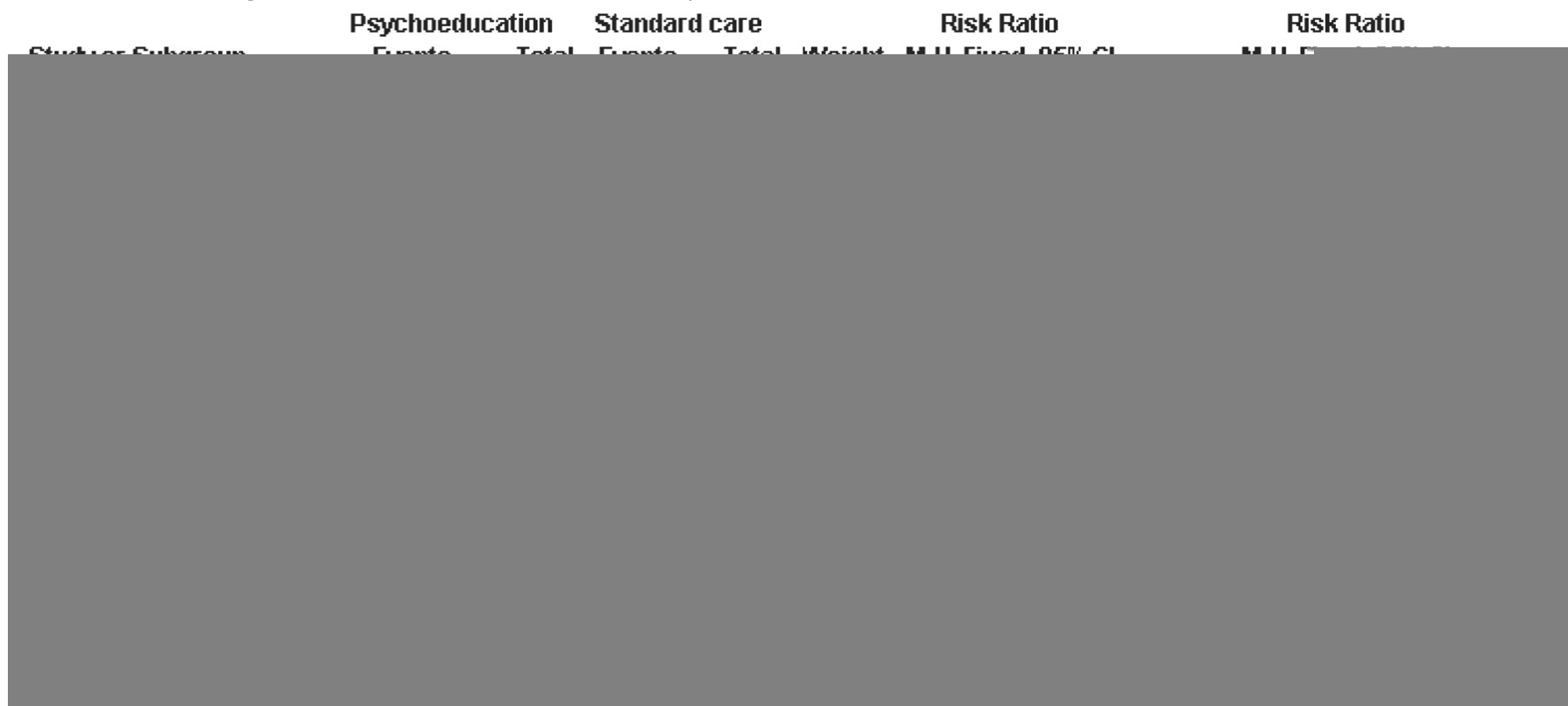
The correct answer is: The effect of both age and year of birth on the cumulative incidence

Question 22

Not answered

Marked out of 1

A meta-analysis has been conducted to identify the effect of psychoeducation in patients with schizophrenia. The results are presented in a graphical format below. (Xia et al. (2011) Schizophrenia Bulletin vol. 37, 21 -22). Answer the following question based on the below graph. Which of the following studies has the least influence on the pooled estimate?



Select one:

- ☐ Standard-Group 1988
- ☐ Brief -Group 1995a

- ☐ Standard-Unclear 1996
- ☐ Unclear -Both 2008
- ☐ Standard-Both 2008 b

Your answer is incorrect.

The study with least weighting will have the least influence on the pooled estimate. The study, unclear-both 2008 has a weight of 0.4% - the least of all.

The correct answer is: Unclear -Both 2008

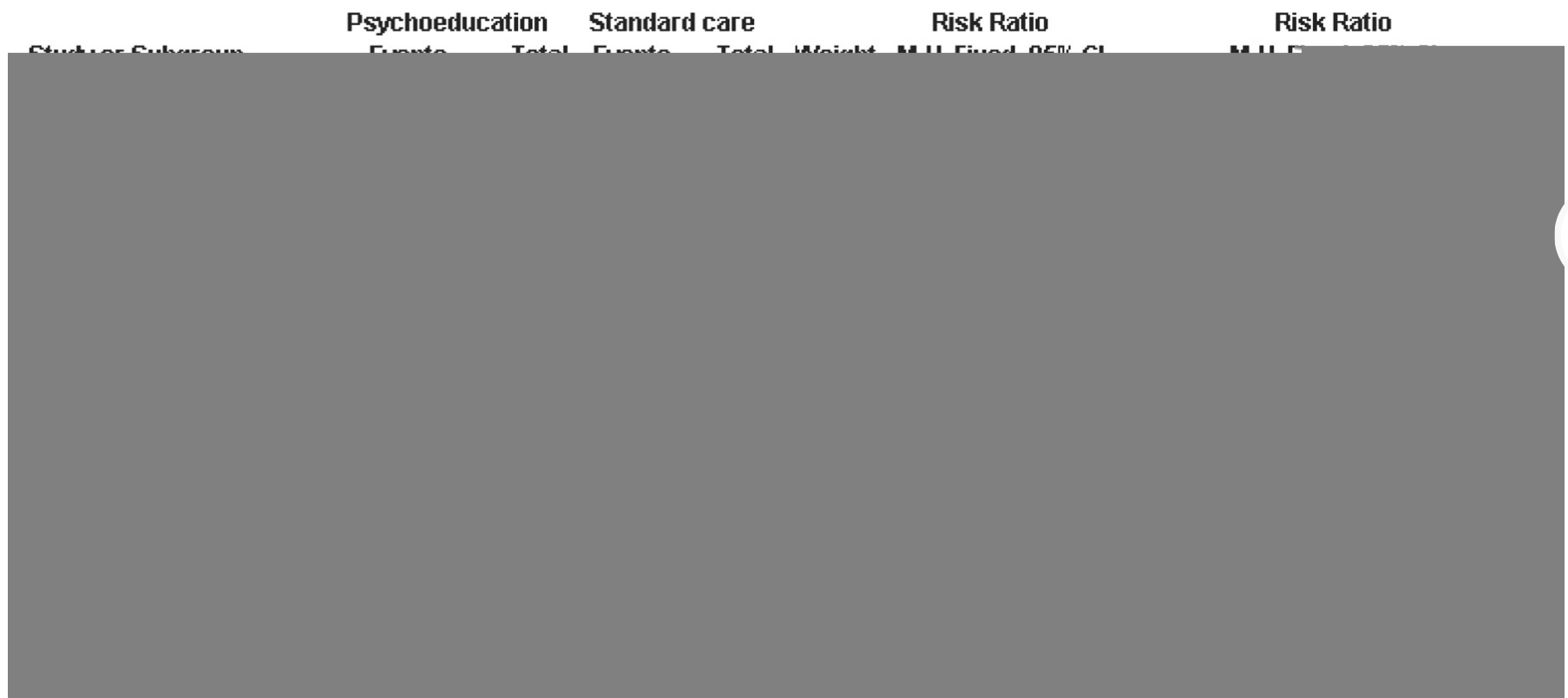


Question 23

Not answered

Marked out of 1

A meta-analysis has been conducted to identify the effect of psychoeducation in patients with schizophrenia. The results are presented in a graphical format below. (Xia et al. (2011) Schizophrenia Bulletin vol. 37, 21 -22). Answer the following question based on the below graph. Some studies in the graph are represented by horizontal lines with an arrowhead. Which of the following interpretation is appropriate in this regard?



Select one:

- ☐ The point estimates from these studies were imprecise

- ☐ These studies were more representative of the target population
- ☐ The follow-up periods for these studies were longer than other studies
- ☐ The point estimates from these studies were of larger magnitude than other studies
- ☐ The sample size for these studies were larger than other studies

Your answer is incorrect.

The length of the horizontal lines reflects the degree of imprecision. In other words, longer the line is, the more imprecise the study is. For some highly imprecise studies, it is not possible to show the full length of the lines in the plot. So arrowheads are used to depict that they extend beyond what has been shown in the forest plot.

The correct answer is: The point estimates from these studies were imprecise

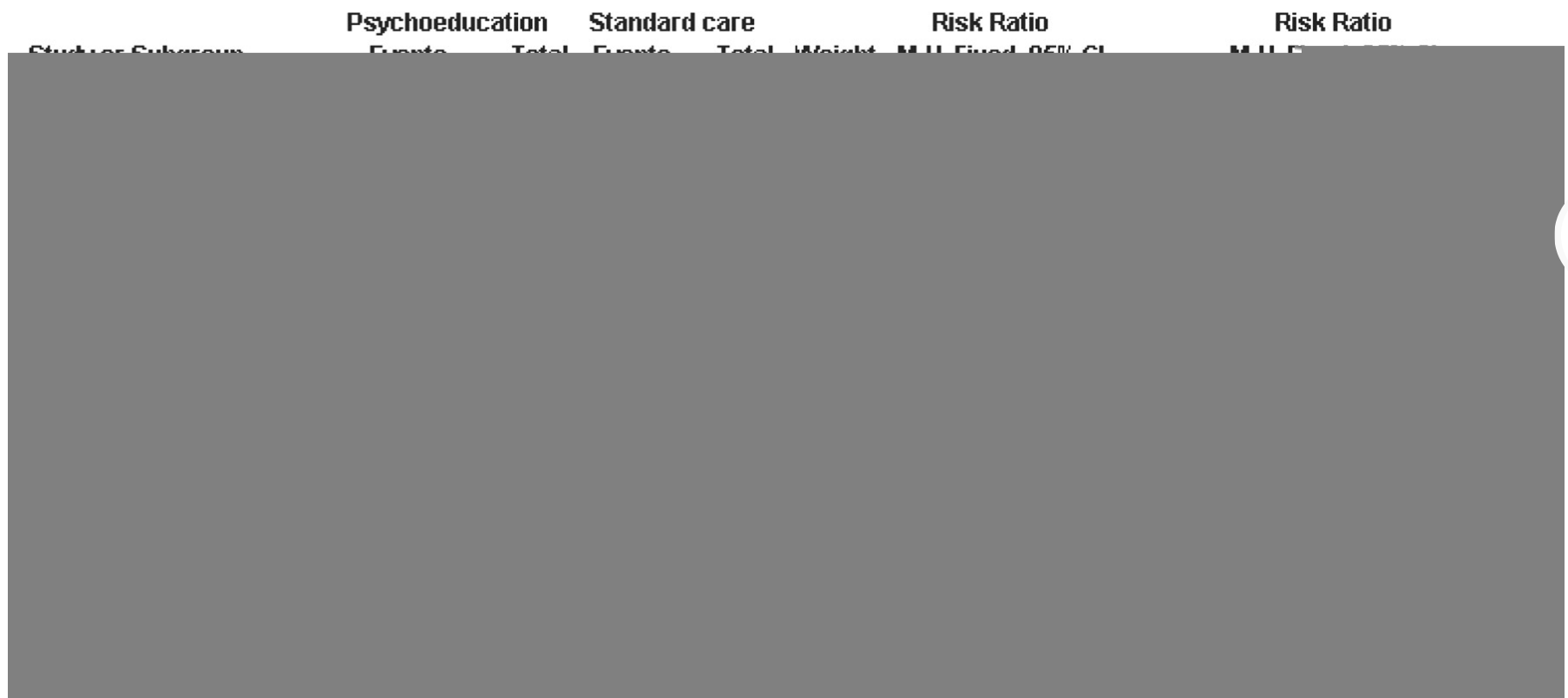


Question **24**

Not answered

Marked out of 1

A meta-analysis has been conducted to identify the effect of psychoeducation in patients with schizophrenia. The results are presented in a graphical format below. (Xia et al. (2011) Schizophrenia Bulletin vol. 37, 21 -22). Answer the following question based on the below graph. The events described in the graph are relapses of psychosis in schizophrenia. The beneficial effect of psychoeducation in the total sample (n=1214) is given by



Select one:

☐ 10.2%

- ☐ 70%
- ☐ 37.9%
- ☐ 27.7%
- ☐ 100%

Your answer is incorrect.

The term beneficial effect is nothing but ABI (Absolute Benefit Increase. $ABI \text{ or } ARR = EER - CER$. From the numbers given in the bottom line (for pooled estimate) $EER = (174/628)$, while $CER = (222/586)$. The $ABI = ((174/628) - (222/586))$. This will be 10.2%

The correct answer is: 10.2%

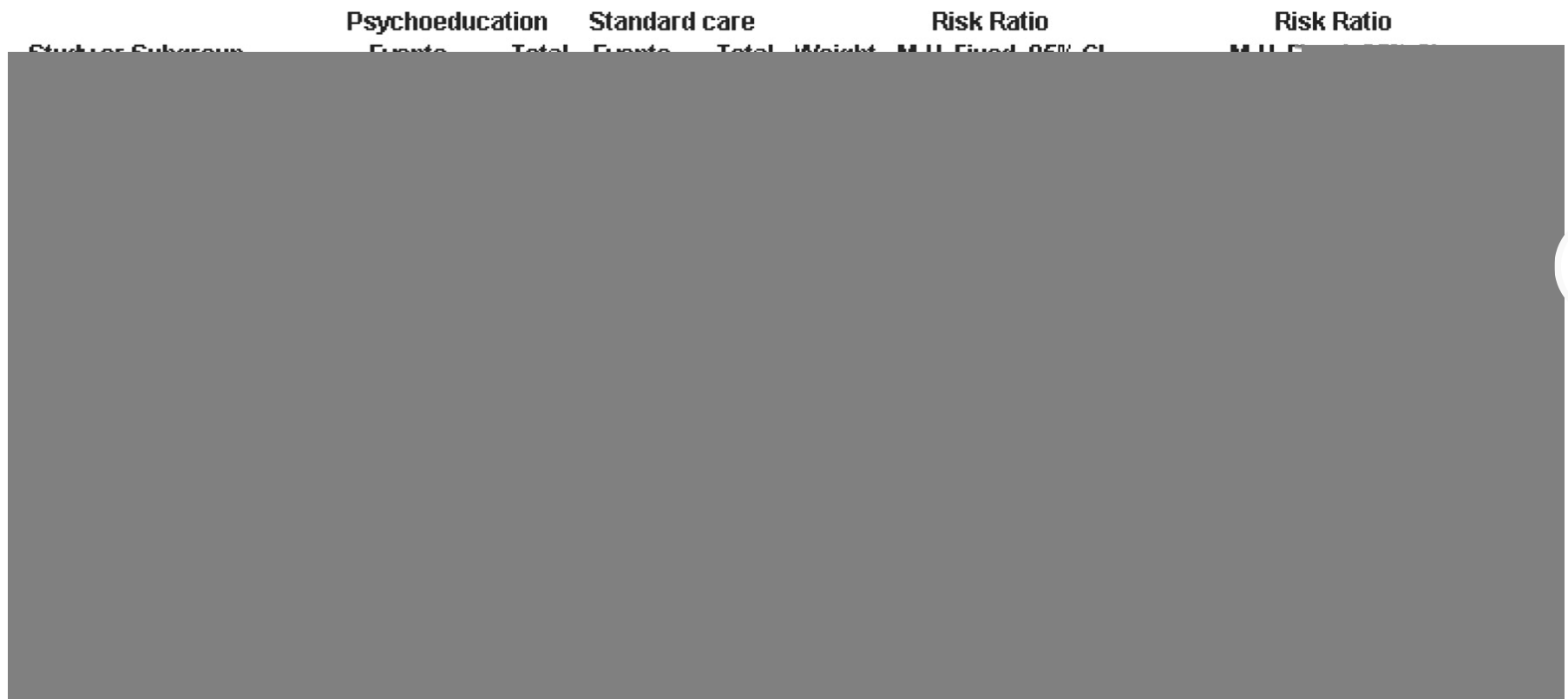


Question 25

Not answered

Marked out of 1

A meta-analysis has been conducted to identify the effect of psychoeducation in patients with schizophrenia. The results are presented in a graphical format below. (Xia et al. (2011) Schizophrenia Bulletin vol. 37, 21 -22). Answer the following question based on the below graph. An individual study with a statistically significant result in favour of psychoeducation includes



Select one:

- ☐ Standard Group 1988
- ☐ Brief -Group 1995a

- ☐ Standard Unclear 2005
- ☐ Brief - Group 1999
- ☐ Standard - Unclear 1996

Your answer is incorrect.

Based on 95% Confidence Intervals reported, Brief-Group 1995a, Standard Both 2008b, Standard Unclear 2006 and Unclear Both 2007, are the trials with statistically significant results.

The correct answer is: Brief -Group 1995a

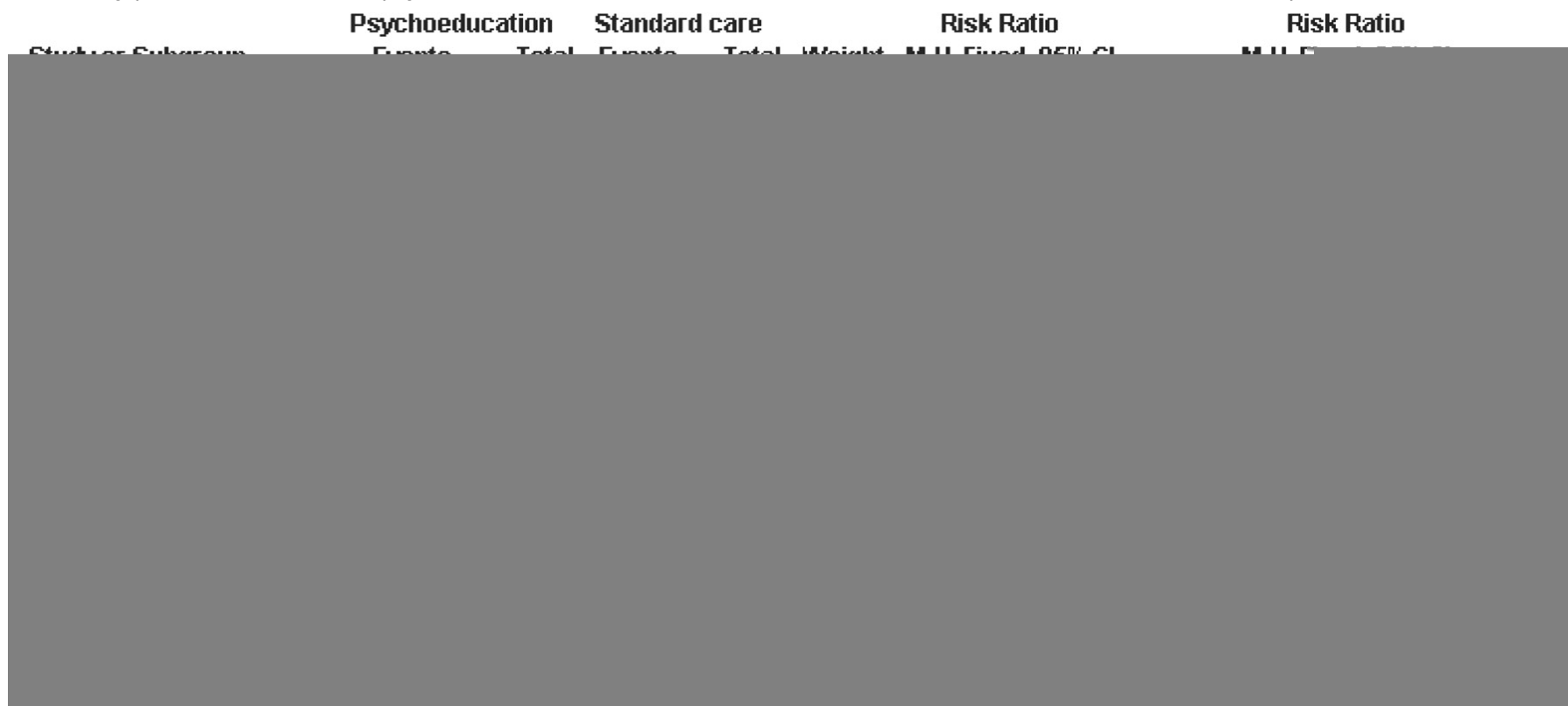


Question 26

Not answered

Marked out of 1

A meta-analysis has been conducted to identify the effect of psychoeducation in patients with schizophrenia. The results are presented in a graphical format below. (Xia et al. (2011) Schizophrenia Bulletin vol. 37, 21 -22). Answer the following question based on the below graph. How many patients must receive psychoeducation rather than standard care as an intervention to reduce one relapse in medium term?



Select one:

☐ 70

☐ 7

- ☐ 2
- ☐ 10
- ☐ 9

Your answer is incorrect.

This is NNT. From the numbers given in the bottom line (for pooled estimate) $EER = (174/628)$, while $CER = (222/586)$. The ABI or ARR = $EER - CER$. $((174/628) - (222/586))$. $ABI = 10.2\%$. Then $NNT = 1/ABI = 1/0.102 = 10$

The correct answer is: 10



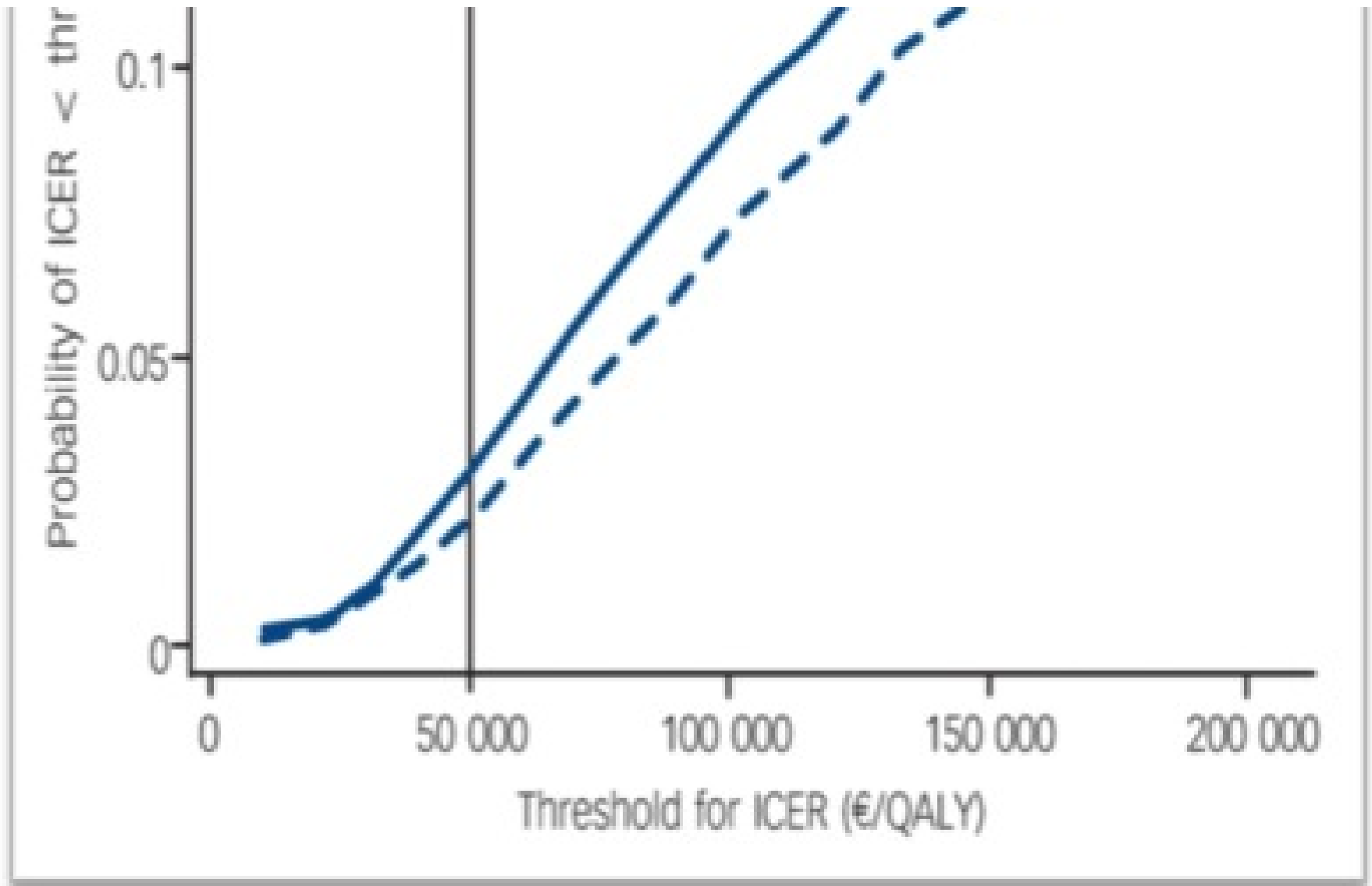
Question 27

Not answered

Marked out of 1

A cluster RCT was conducted in Germany to analyse the cost-effectiveness of an optimised care model for people with anxiety disorders in primary care (Konig, H.-H. et al. The British Journal of Psychiatry 2009;195:308-317). 46 primary care practices with 389 individuals with anxiety were randomised to intervention (23 practices, 201 participants) or usual care (23 practices, 188 participants). For monetary valuation of resource utilisation, unit costs were determined for all services used and for all medical goods prescribed or privately purchased at 2006 price levels. Informal care provided by relatives and friends was valued by the mean net income per h (Euros 18.30) in Germany. Productivity loss was quantified based on the human capital approach, valuing illness-related work loss days by the mean labour costs per working day (Euros 169) in Germany. A generic health-related quality of life questionnaire called EQ-5D was used for primary outcome determination. It comprises five questions (items) relating to current problems in the dimensions: mobility; self-care; usual activities; pain/discomfort; and anxiety/depression. Quality-adjusted life-years were calculated by weighting the duration of health states by the EQ-5D index. Beck's anxiety inventory was also used to quantify clinical improvement. To analyse the impact of various assumptions made when calculating QALYs and costs, analyses were performed by varying the following parameters: 1. Using German version of EQ-5D instead or British version 2. Using a different scale called EQ-VAS as QALY weights. 3. Calculating indirect costs using a different approach 4. Varying intervention costs by plus and minus 100%. 5. Calculating ICER for only complete cases with no missing values instead of including all in the analysis. Total costs were higher in the intervention group (Euros 4911 v. Euros 3453, $P = 0.09$). Mean incremental total costs were Euros 1459. Mean incremental QALYs were -0.0042. Mean difference in BAI score at initial follow-up was 3 points, with intervention group doing better than control group, though this result was not significant. The probability of an incremental cost-effectiveness ratio < Euros 50 000 per quality-adjusted life year was below 10%. What is the below graph called?





Select one:

☐ Scatter plot

- ☐ Cost effectiveness acceptability curve
- ☐ Kaplan Meier curve
- ☐ Blobbogram
- ☐ Cost effectiveness plane

Your answer is incorrect.

The attached graph is a cost-effectiveness acceptability curve. It is a plot between incremental cost-effectiveness threshold (or willingness to pay) and the probability that the ICER is less than the threshold set (i.e. the chances of cost-effectiveness).

The correct answer is: Cost effectiveness acceptability curve



Question 28

Not answered

Marked out of 1

A cluster RCT was conducted in Germany to analyse the cost-effectiveness of an optimised care model for people with anxiety disorders in primary care (Konig, H.-H. et al. The British Journal of Psychiatry 2009;195:308-317). 46 primary care practices with 389 individuals with anxiety were randomised to intervention (23 practices, 201 participants) or usual care (23 practices, 188 participants). For monetary valuation of resource utilisation, unit costs were determined for all services used and for all medical goods prescribed or privately purchased at 2006 price levels. Informal care provided by relatives and friends was valued by the mean net income per h (Euros 18.30) in Germany. Productivity loss was quantified based on the human capital approach, valuing illness-related work loss days by the mean labour costs per working day (Euros 169) in Germany. A generic health-related quality of life questionnaire called EQ-5D was used for primary outcome determination. It comprises five questions (items) relating to current problems in the dimensions: mobility; self-care; usual activities; pain/discomfort; and anxiety/depression. Quality-adjusted life-years were calculated by weighting the duration of health states by the EQ -5D index. Beck's anxiety inventory was also used to quantify clinical improvement. To analyse the impact of various assumptions made when calculating QALYs and costs, analyses were performed by varying the following parameters: 1. Using German version of EQ-5D instead or British version 2. Using a different scale called EQ -VAS as QALY weights. 3. Calculating indirect costs using a different approach 4. Varying intervention costs by plus and minus 100%. 5. Calculating ICER for only complete cases with no missing values instead of including all in the analysis. Total costs were higher in the intervention group (Euros 4911 v. Euros 3453, P = 0.09). Mean incremental total costs were Euros 1459. Mean incremental QALYs were -0.0042. Mean difference in BAI score at initial follow-up was 3 points, with intervention group doing better than control group, though this result was not significant. The probability of an incremental cost-effectiveness ratio < Euros 50 000 per quality-adjusted life year was below 10%. The below graph is used to

Select one:

- ☐ Estimate the confidence limits around mean incremental cost effectiveness ratio
- ☐ Differentiate direct from indirect costs in an economic analysis
- ☐ Estimate incremental cost effectiveness ratio
- ☐ Study the effect of attrition rate on incremental cost effectiveness ratio
- ☐ Visualize statistical uncertainty of the incremental cost effectiveness ratio

Your answer is incorrect.

The probability of ICER is nothing but a measure of statistical uncertainty over the cost-effectiveness projections. The above plot is helpful to visualise this uncertainty in the context of the available resources.

The correct answer is: Visualize statistical uncertainty of the incremental cost effectiveness ratio



Question 29

Not answered

Marked out of 1

A cluster RCT was conducted in Germany to analyse the cost-effectiveness of an optimised care model for people with anxiety disorders in primary care (Konig, H.-H. et al. The British Journal of Psychiatry 2009;195:308-317). 46 primary care practices with 389 individuals with anxiety were randomised to intervention (23 practices, 201 participants) or usual care (23 practices, 188 participants). For monetary valuation of resource utilisation, unit costs were determined for all services used and for all medical goods prescribed or privately purchased at 2006 price levels. Informal care provided by relatives and friends was valued by the mean net income per h (Euros 18.30) in Germany. Productivity loss was quantified based on the human capital approach, valuing illness-related work loss days by the mean labour costs per working day (Euros 169) in Germany. A generic health-related quality of life questionnaire called EQ-5D was used for primary outcome determination. It comprises five questions (items) relating to current problems in the dimensions: mobility; self-care; usual activities; pain/discomfort; and anxiety/depression. Quality-adjusted life-years were calculated by weighting the duration of health states by the EQ -5D index. Beck's anxiety inventory was also used to quantify clinical improvement. To analyse the impact of various assumptions made when calculating QALYs and costs, analyses were performed by varying the following parameters: 1. Using German version of EQ-5D instead or British version 2. Using a different scale called EQ -VAS as QALY weights. 3. Calculating indirect costs using a different approach 4. Varying intervention costs by plus and minus 100%. 5. Calculating ICER for only complete cases with no missing values instead of including all in the analysis. Total costs were higher in the intervention group (Euros 4911 v. Euros 3453, $P = 0.09$). Mean incremental total costs were Euros 1459. Mean incremental QALYs were -0.0042. Mean difference in BAI score at initial follow-up was 3 points, with intervention group doing better than control group, though this result was not significant. The probability of an incremental cost-effectiveness ratio < Euros 50 000 per quality-adjusted life year was below 10%. With respect to the primary outcome, what type of study has been conducted?

Select one:

- ☐ Cost benefit study
- ☐ Cost trade-off study
- ☐ Cost minimization study
- ☐ Cost utility study
- ☐ Cost effectiveness study

Your answer is incorrect.

In this study, QALYs are used as primary outcome measures. This is best termed as a cost-utility study.

The correct answer is: Cost utility study



Question 30

Not answered

Marked out of 1

A cluster RCT was conducted in Germany to analyse the cost-effectiveness of an optimised care model for people with anxiety disorders in primary care (Konig, H.-H. et al. The British Journal of Psychiatry 2009;195:308-317). 46 primary care practices with 389 individuals with anxiety were randomised to intervention (23 practices, 201 participants) or usual care (23 practices, 188 participants). For monetary valuation of resource utilisation, unit costs were determined for all services used and for all medical goods prescribed or privately purchased at 2006 price levels. Informal care provided by relatives and friends was valued by the mean net income per h (Euros 18.30) in Germany. Productivity loss was quantified based on the human capital approach, valuing illness-related work loss days by the mean labour costs per working day (Euros 169) in Germany. A generic health-related quality of life questionnaire called EQ-5D was used for primary outcome determination. It comprises five questions (items) relating to current problems in the dimensions: mobility; self-care; usual activities; pain/discomfort; and anxiety/depression. Quality-adjusted life-years were calculated by weighting the duration of health states by the EQ -5D index. Beck's anxiety inventory was also used to quantify clinical improvement. To analyse the impact of various assumptions made when calculating QALYs and costs, analyses were performed by varying the following parameters: 1. Using German version of EQ-5D instead or British version 2. Using a different scale called EQ -VAS as QALY weights. 3. Calculating indirect costs using a different approach 4. Varying intervention costs by plus and minus 100%. 5. Calculating ICER for only complete cases with no missing values instead of including all in the analysis. Total costs were higher in the intervention group (Euros 4911 v. Euros 3453, $P = 0.09$). Mean incremental total costs were Euros 1459. Mean incremental QALYs were -0.0042. Mean difference in BAI score at initial follow-up was 3 points, with intervention group doing better than control group, though this result was not significant. The probability of an incremental cost-effectiveness ratio < Euros 50 000 per quality-adjusted life year was below 10%. Productivity loss was quantified to be included in the cost estimates. Which of the following category of costs does productivity loss belong to?

Select one:

- ☐ Intangible costs
- ☐ Indirect costs
- ☐ Direct costs
- ☐ Discount costs
- ☐ Inflation costs

Your answer is incorrect.

Productivity loss is a cost borne by the public, not the health system. Hence it is best termed as an indirect cost.

The correct answer is: Indirect costs



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